

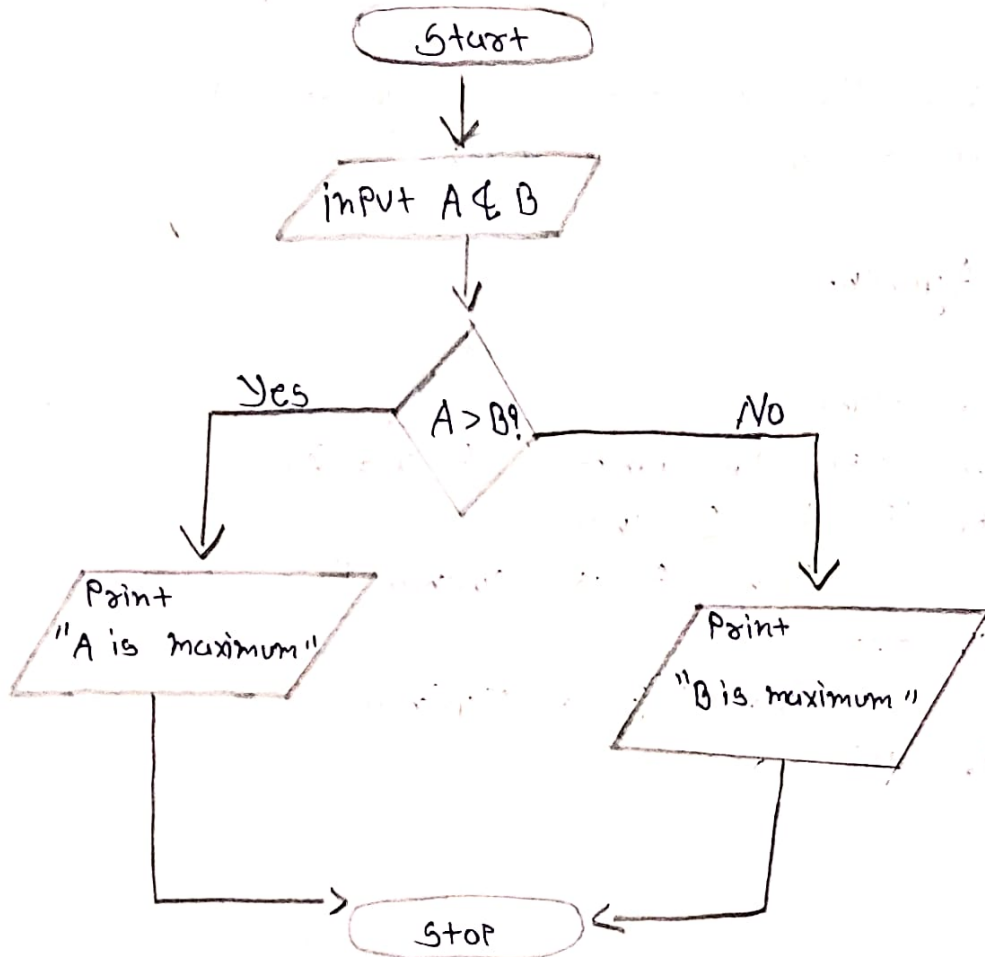
Prepare flowchart & algorithm for a given problem.

Q.1 Draw flowchart and write algorithm to find a maximum out of two given numbers.

→ Algorithm

1. Start
2. Input two numbers A and B.
3. if $A > B$ then
 Print "A is Maximum"
4. else
 Print "B is Maximum"
5. stop

Flow Chart :-



Q.2 Draw Flowchart and write Algorithm to find whether a given number is odd or even.

→ Algorithm

1. Start

2. Input N

3. Divide by 2

4. if remainder $== 0$

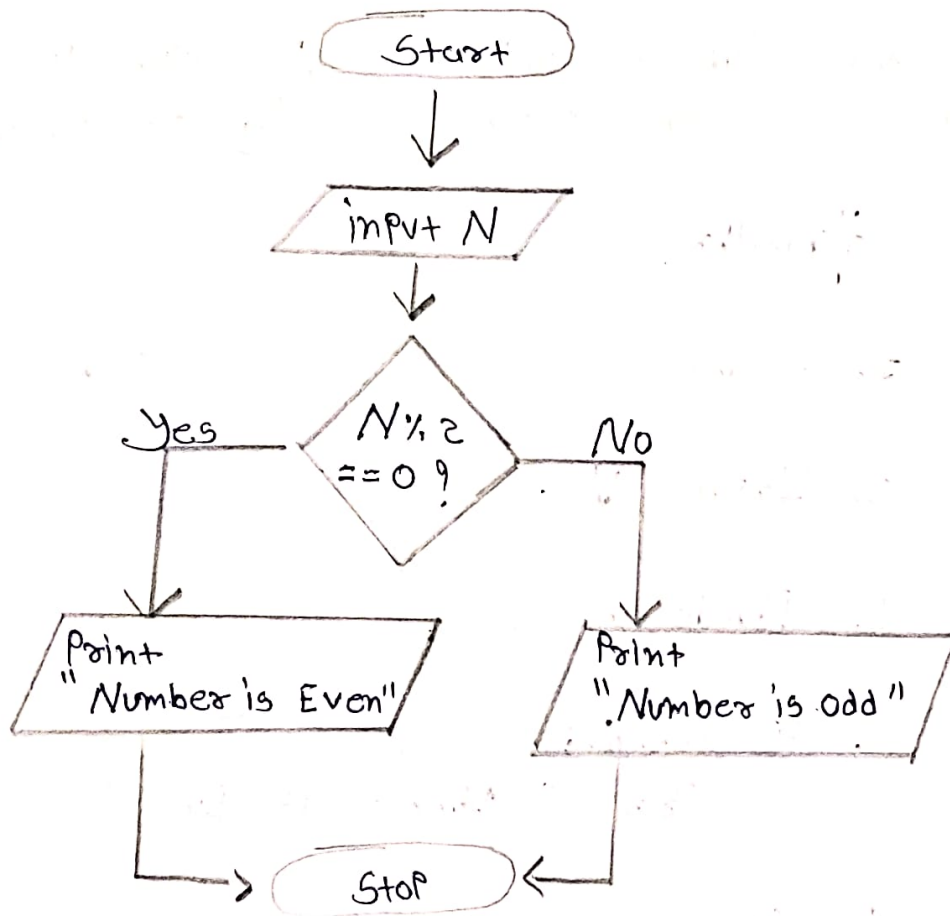
Print "Number is Even"

5. else

Print "Number is odd"

6. Stop

Flow Chart :-



EXPT. NO.	NAME	M T W T F S S	YOUVA
		Page No.	
		Date	

Q.3 Draw Flowchart and write Algorithm to check whether the given number is positive, negative or zero.

→ Algorithm

1. Start

2. input N

3. if $N > 0$ then

Print "Number is Positive"

4. elif $N == 0$ then

Print "Number is zero"

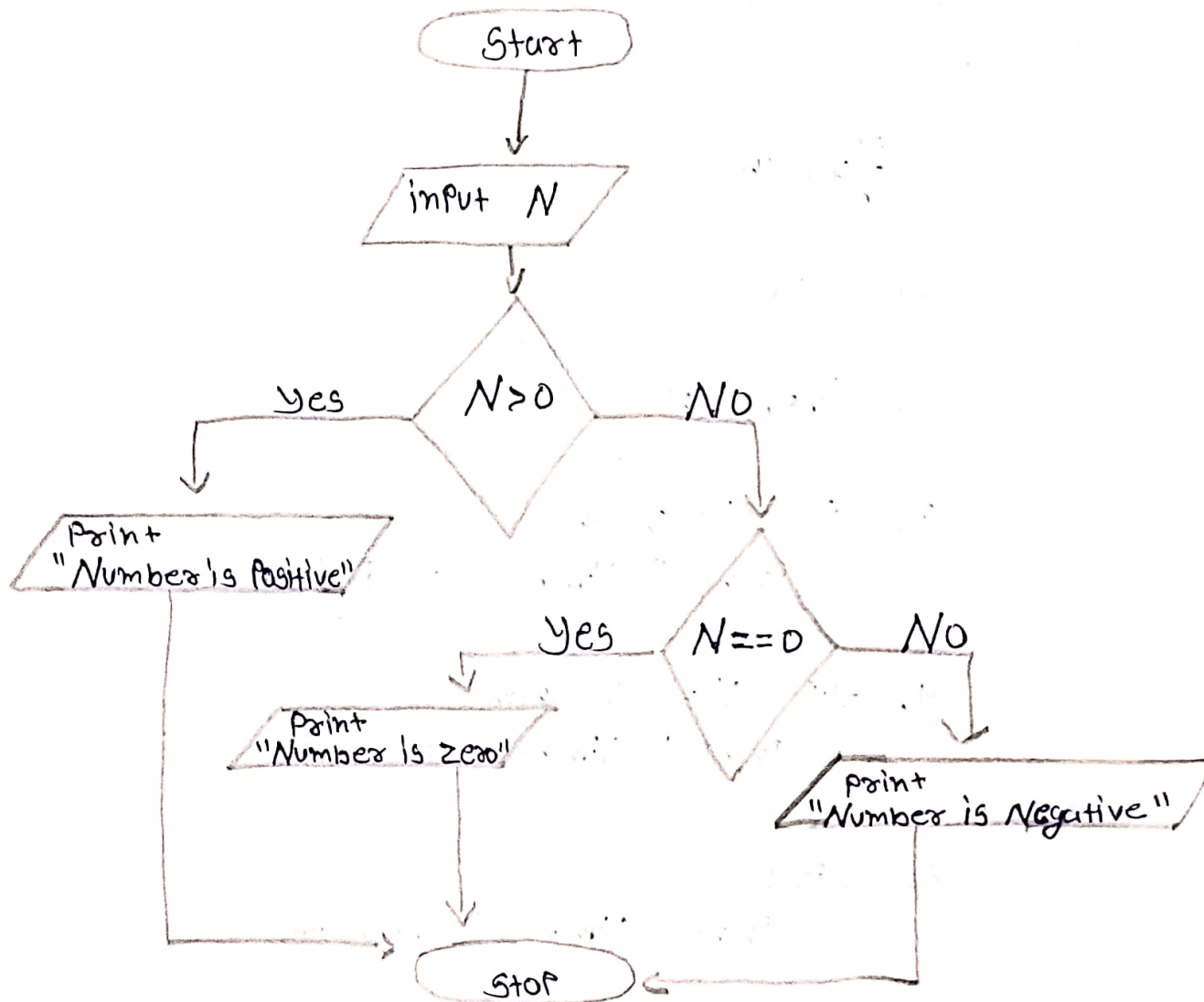
5. else:

Print "Number is Negative"

6. STOP

Teacher's Signature: _____

Flow Chart



EXPT.
NO.

NAME

Page No.

Date

YOUVA

1.

(i) Find Sum of two given numbers.

→ Algorithm

1. Start
2. Input A und B
3. $Sum = A + B$
4. Display Sum
5. Stop

ii) Find a maximum out of two given numbers.

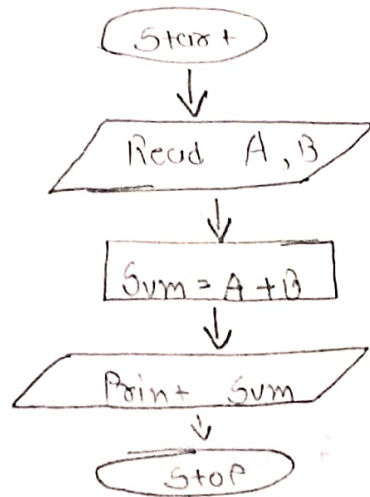
→ Algorithm

1. Start
2. Input A und B
3. If $A > B$ Print A
4. Else Print B
5. Stop

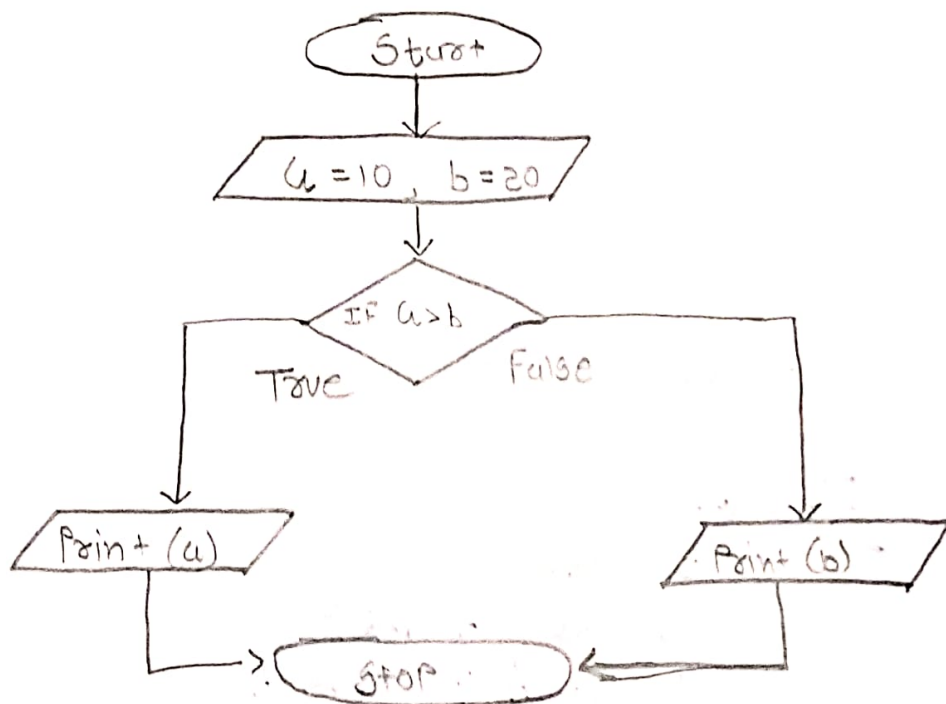
Teacher's Signature

★ Flowchart :-

(i)



(ii)



EXPT. NO.	NAME	M T W T F S S						
		Page No.:						
		Date:						
		YOUVA						

iii) Find whether a given number is odd or even.

→

Algorithm

1. Start
2. Input number N
3. If $N \% 2 == 0$ Print Even
4. Else print Odd
5. Stop

iv) Find a maximum out of three given numbers.

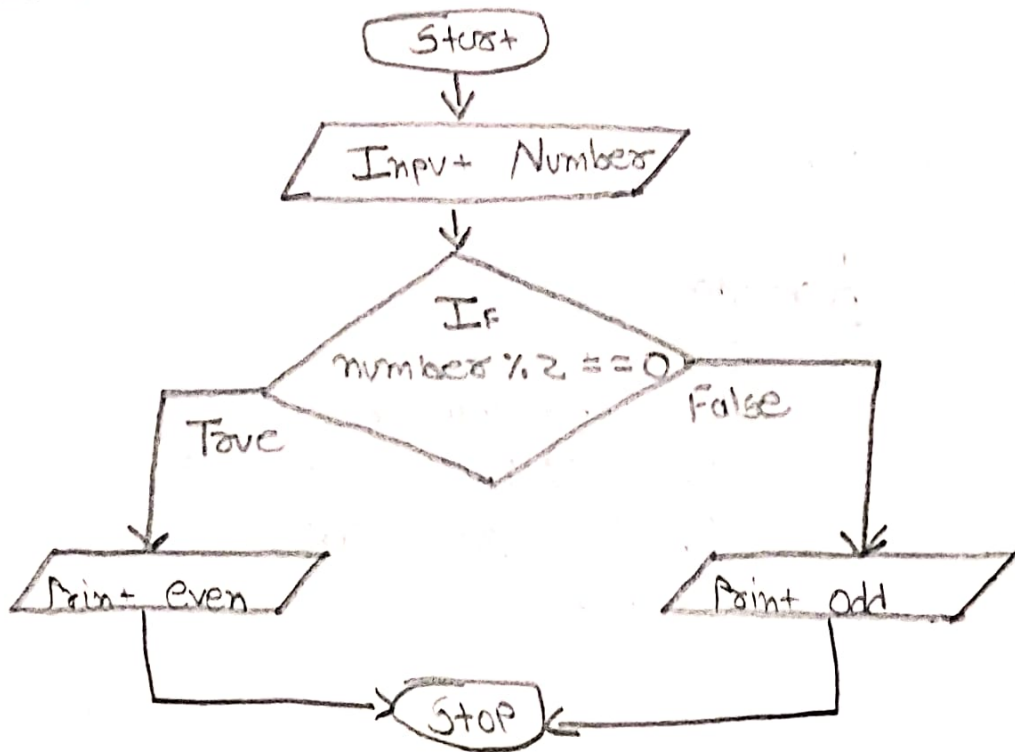
→

Algorithm

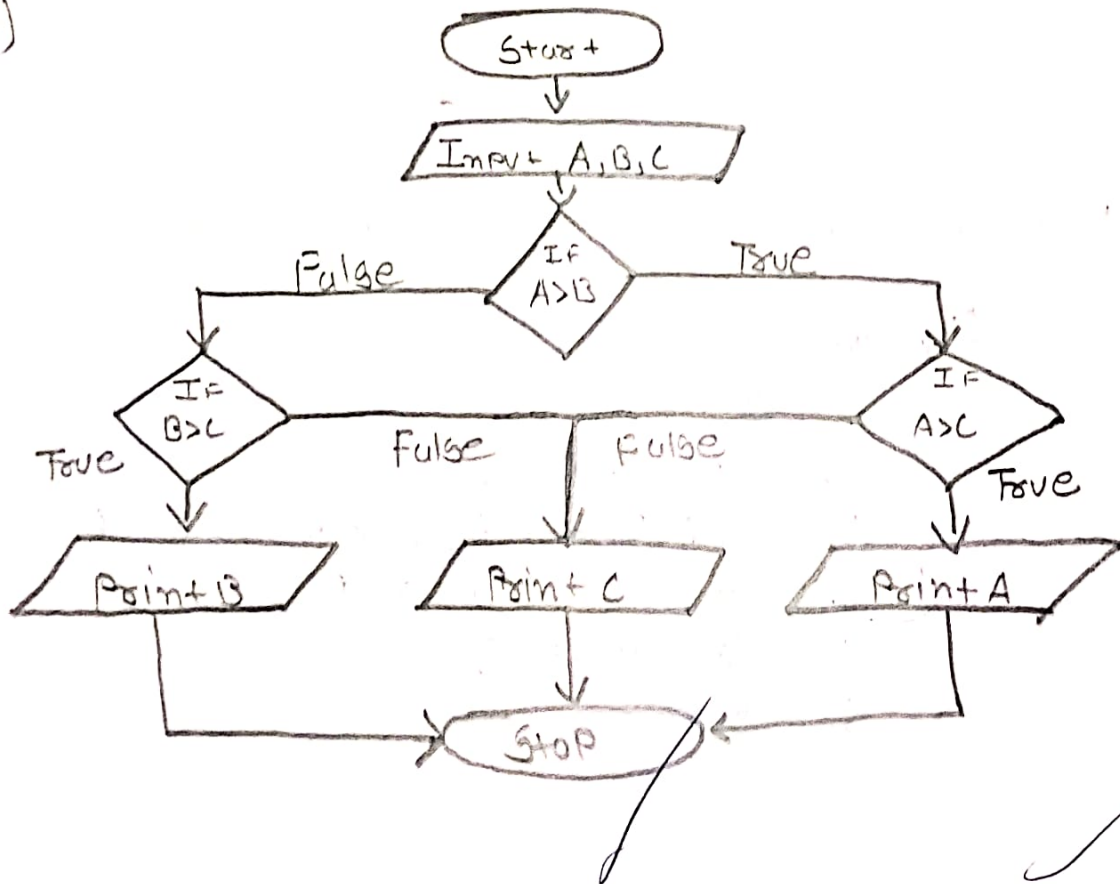
1. Start
2. Input A, B & C
3. If $A > B$ and $A > C$ then print A
4. Else if $B > C$ Print B
5. Else print C
6. End

Teacher's Signature: _____

★ Flowchart :-
(iii)



(iv)



EXPT NO.	NAME	M T W T F S S						
		Page No.						
		Date						
		YOUVA						

2.

a) Write a Program to Print your name, mobile number, and date of birth. (Flowchart & Algorithm.)

→ Algorithm

1. Start

2. Print Name

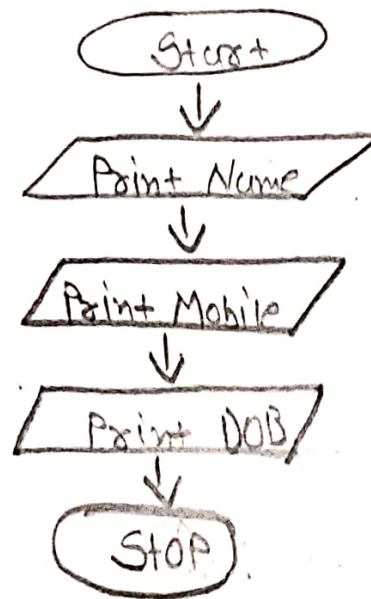
3. Print Mobile Number

4. Print DOB

5. END

Teacher's Signature: _____

★ Flowchart :-

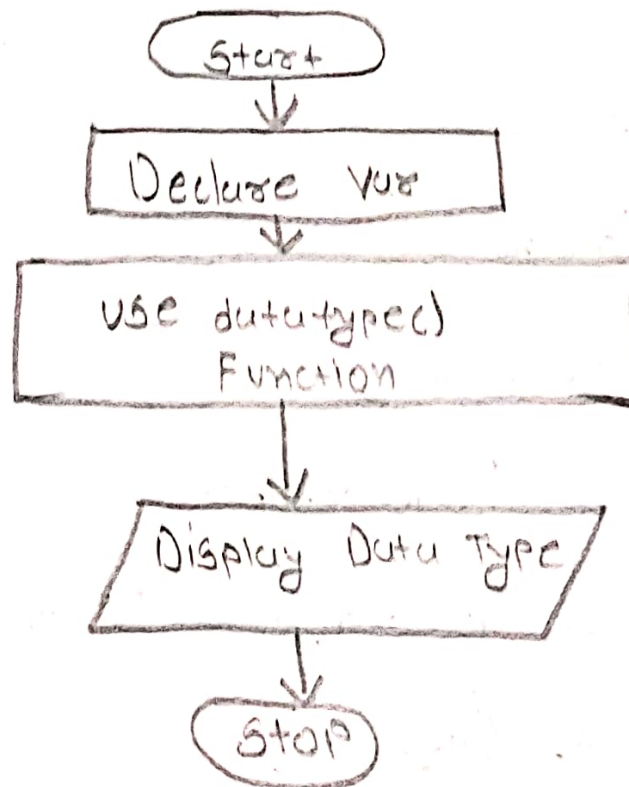


b) Develop a program to identify data-types in Python.
(Algorithm & Flowchart)

→ Algorithm

1. Start
2. Declare variables
3. Use data type() Function
4. Display datatype
5. Stop

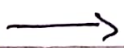
★ Flowchart :-



EXPT. NO.	NAME	M T W T F S S	YOUVA
		Page No.	
		Date	

3.

a) Create a program to read three numbers from the User and find the average of the numbers.



Algorithm

1. Start

2. Input A, B & C

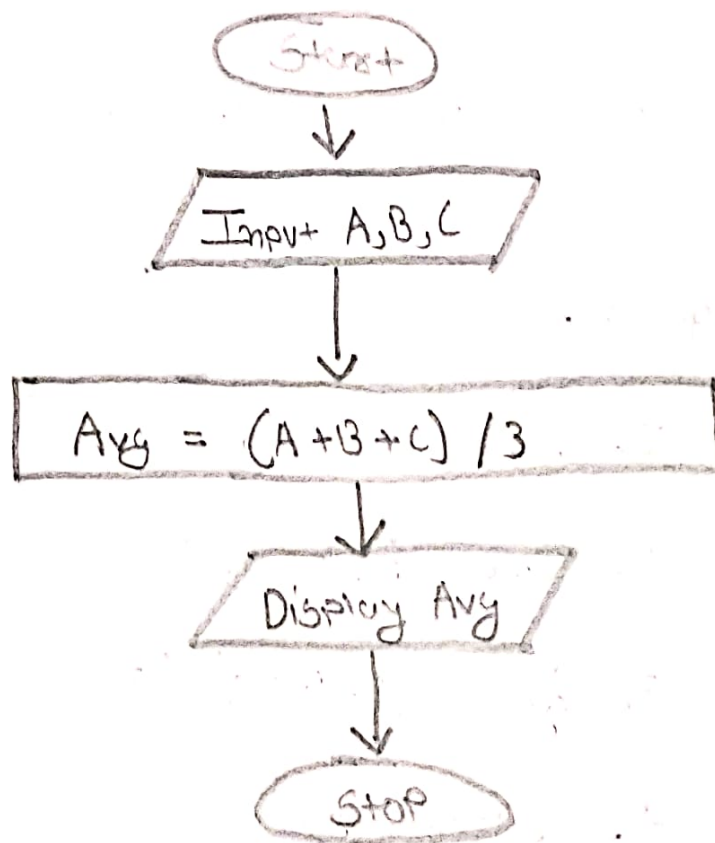
3. $Avg = (A+B+C) / 3$

4. Display Avg

5. Stop

Teacher's Signature: _____

★ Flowchart :-



EXPT. NO.	NAME	M T W T F S S
		Page No.
		Date
		YOUVA

b) Develop a program that can calculate simple interest and compound interest on given data.

→ Algorithm

1. Start

2. Input P, R, T

3. $SI = (P \times R \times T) / 100$

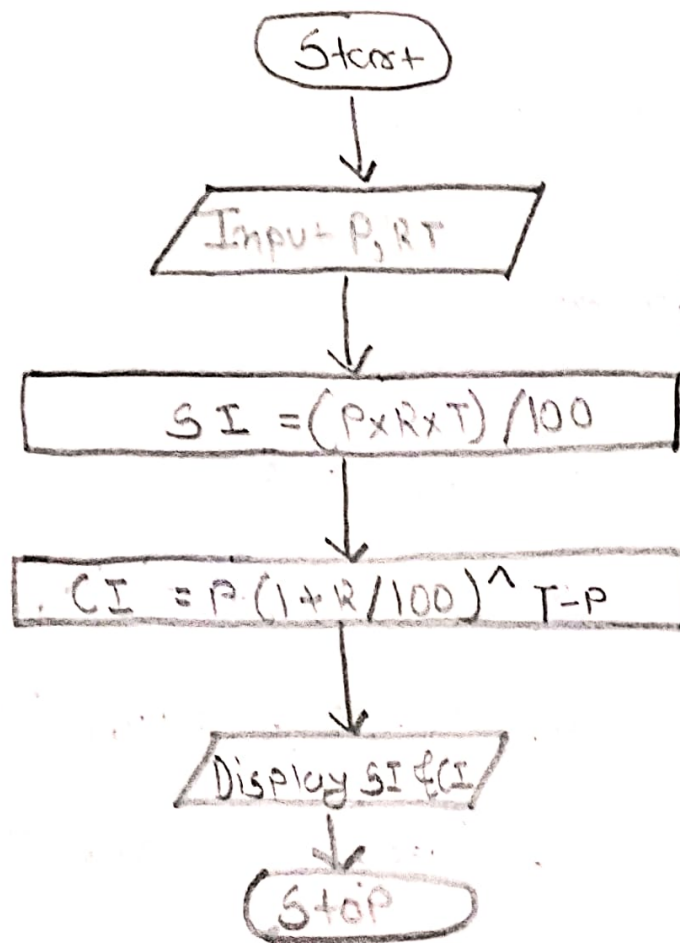
4. $CI = P(1 + R/100)^T - P$

5. Display SI and CI

6. Stop

Teacher's Signature

★ Flowchart :-



EXPT. NO.	NAME	M T W T F S S	Page No.	YOUVA
		Date		

Q) Write a program to Convert temperature from Fahrenheit to Celsius unit using eq: $C = (F - 32) / 1.8$

→

Algorithm

1. Start

2. Input Fahrenheit

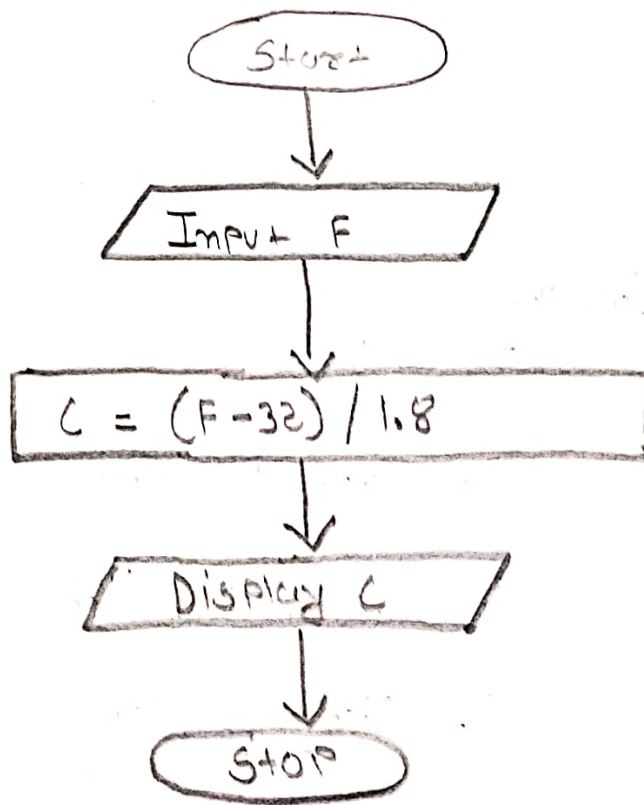
3. $C = (F - 32) / 1.8$

4. Display Celsius

5. Stop

Teacher's Signature

★ Flowchart :-



EXPT. NO.	NAME	M T W T F S S	Page No.	YOUVA
		Date		

4.

a) Identify Whether the scanned number is even or odd and print an appropriate message.

→ Algorithm

1. Start

2. Input N

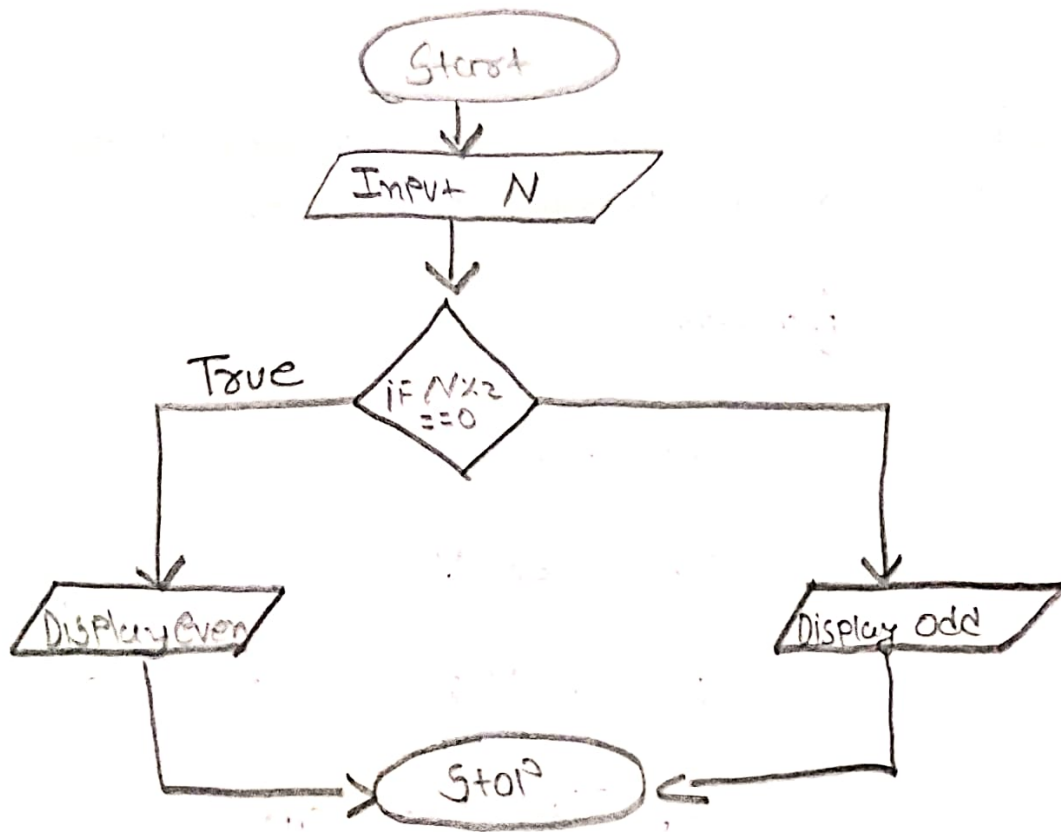
3. IF $N \% 2 == 0$ print Even

4. Else print odd

5. Stop

Teacher's Signature: _____

★ Flowchart :-



EXPT. NO.	NAME	M T W T F S S Page No.	YOUVA
		Date	

b. Create a program to find a maximum number among the given three numbers.

→ Algorithm

1. Start

2. Input A, B & C

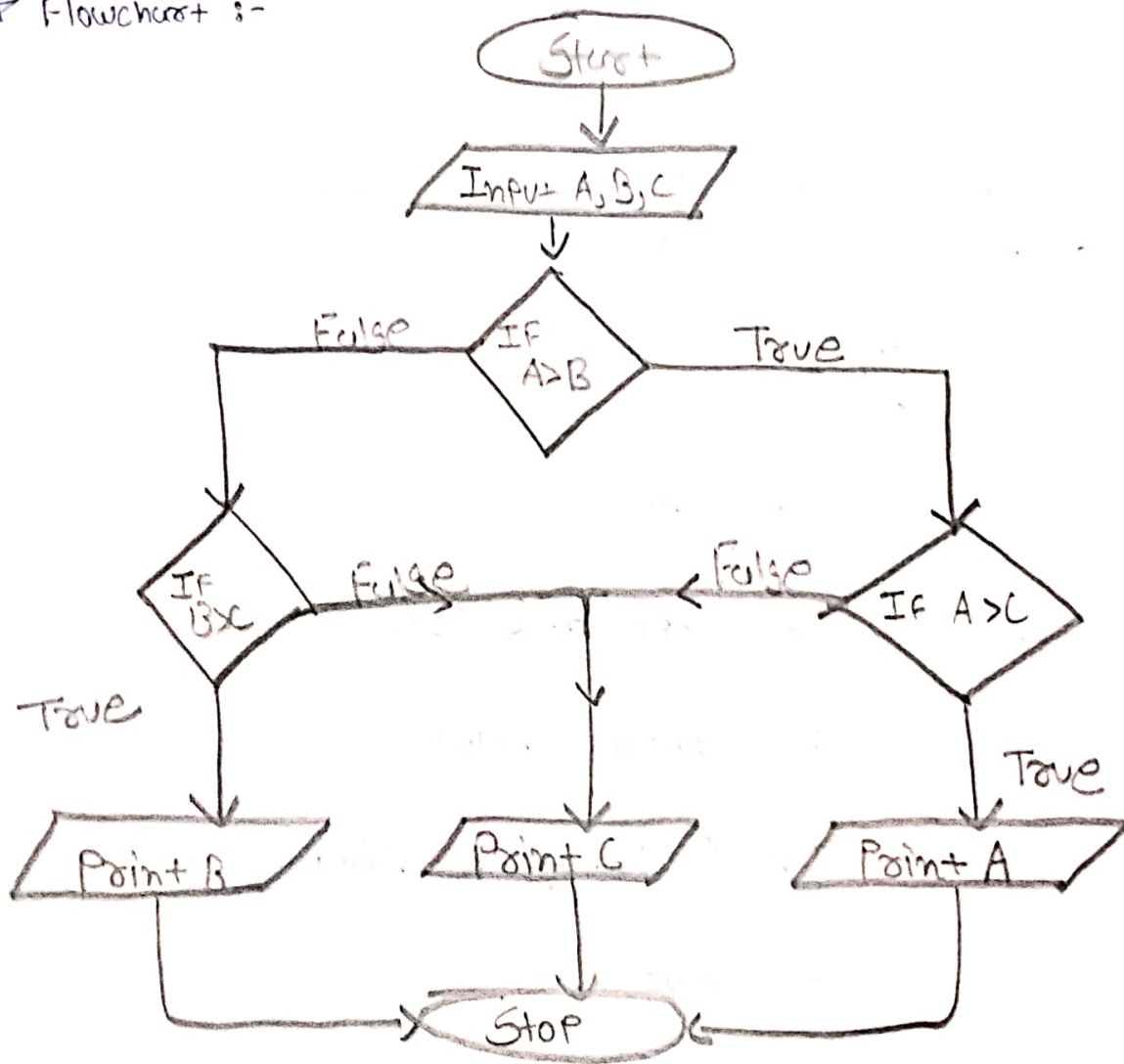
3. Compare Numbers

4. Print largest Number

5. Stop

Teacher's Signature: _____

★ Flowchart :-



EXPT. NO.	NAME	M T W T F S S	YOUVA
		Page No.	Date

5.

c) Develop a program to demonstrate the basic functionalities of a standard calculator.

→ Algorithm

1. Start

2. Input two numbers

3. Input Operator

4. Perform operation using if-elif

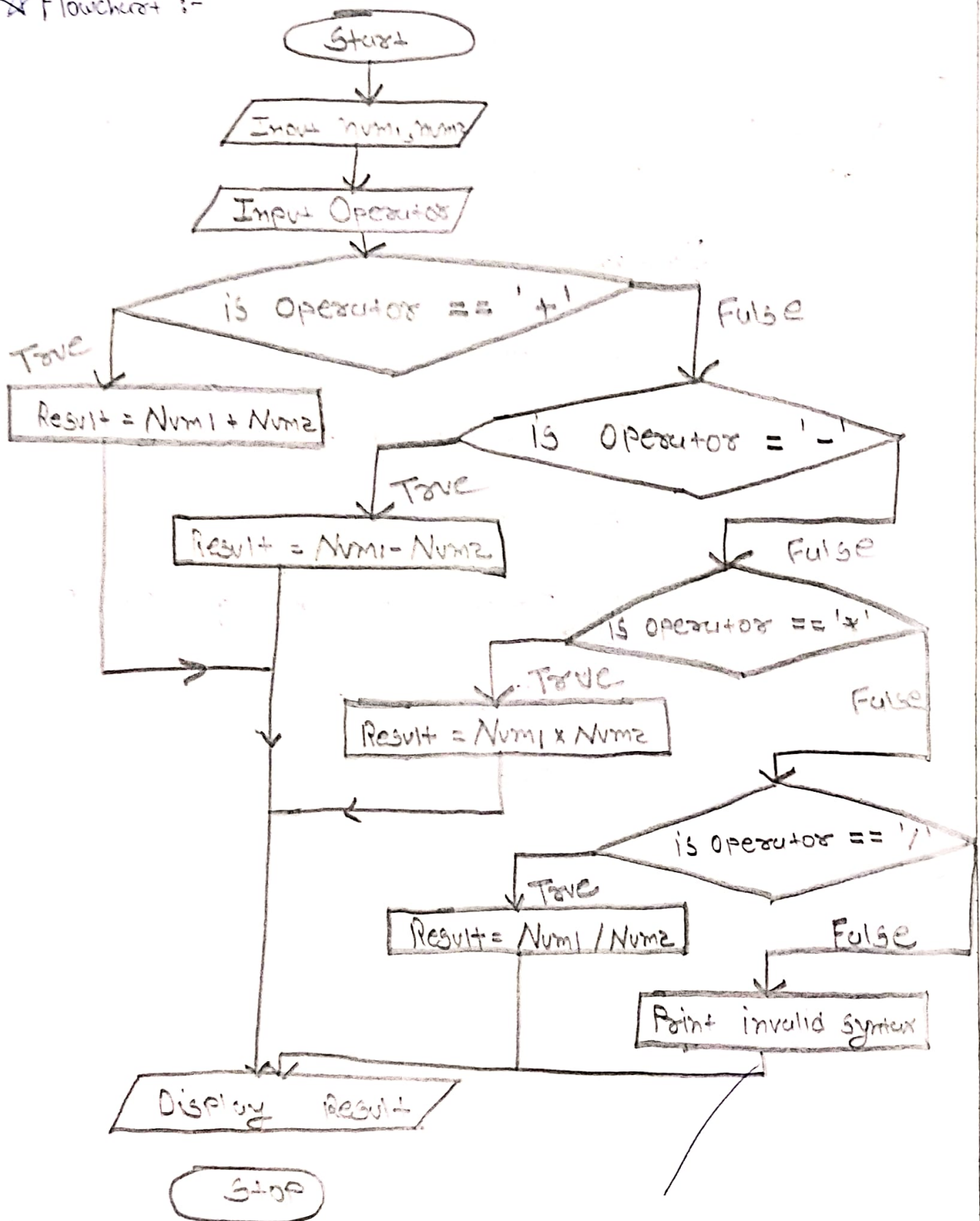
5. Display result

6. End



Teacher's Signature _____

★ Flowchart :-



EXPT. NO.	NAME	M T W T F S S	Page No.	YOUVA
			Date:	

* Draw Flowchart and Write algorithm & Flowchart to Swap two number using temporary variable

→ Algorithm

1. Start

2. Input two number A & B

3. Set temp = A

4. Set A = B

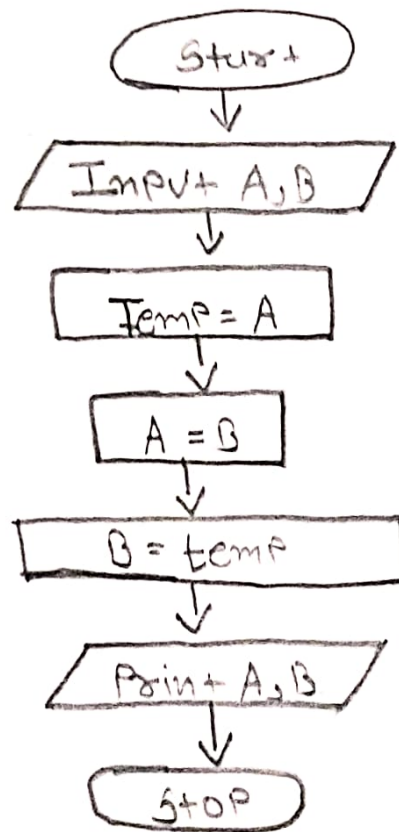
5. Set B = temp

6. Print A, B

7. Stop

Teacher's Signature: _____

★ Flowchart :-



EXPT. NO.	NAME	M T W T F S S	YOUVA
		Page No.	
		Date	

★ Draw Flowchart and write algorithm to convert Temperature from Fahrenheit to Celsius.

→ Algorithm

1. Start

2. Input tem. in Fahrenheit (F)

3. Set $C = (5/9) \times (F - 32)$

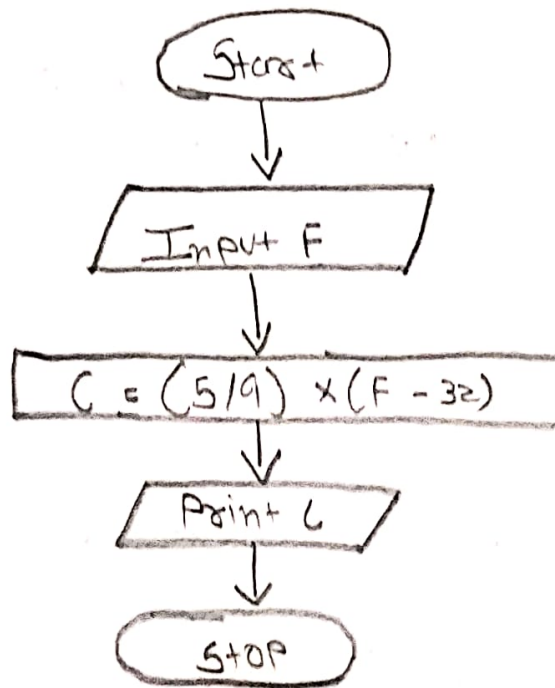
4. Print C

5. Stop

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Teacher's Signature _____

★ Flowchart :-



EXPT. NO.	NAME	M T W T F S S						
		Page No.		YOUVA				
		Date						

★ Draw Flowchart and write algorithm to find Even number between 1 to 50.

→ Algorithm

1. Start

2. Set $N = 2$

3. Check if $N \leq 50$

4. If yes, Print N

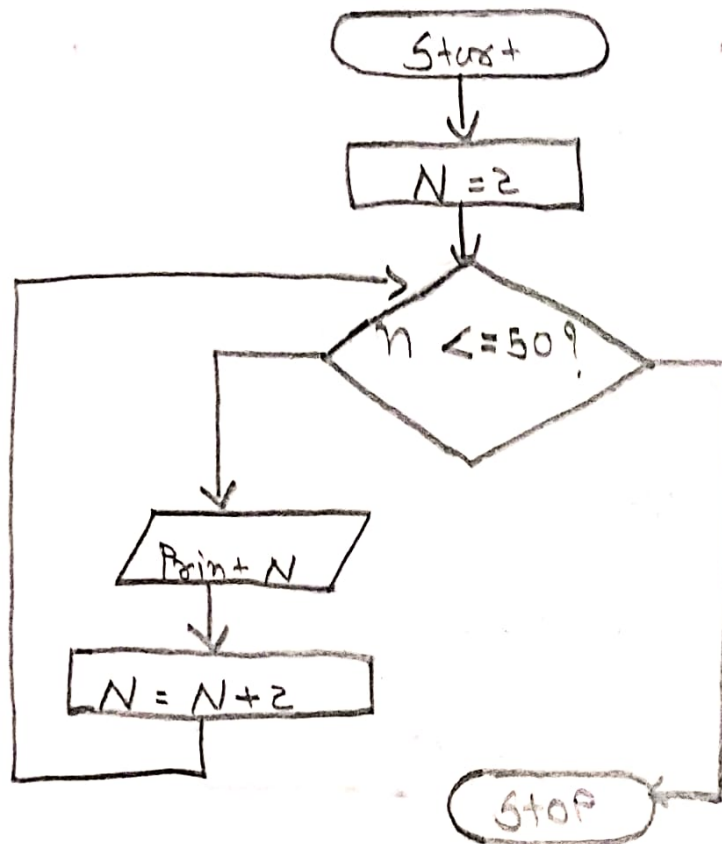
5. set $N = N + 2$

6. Go to step 3

7. Stop.

Teacher's Signature: _____

★ Flowchart :-



EXPT. NO.	NAME	M T W T F S S	Page No.	YOUVA
		Date		

6.

a) Write a Python Program to print (1 - 10) numbers using loops. (flowchart & algorithm)

—> Algorithm

1. Start

2. Initialize $i = 1$

3. Check Condition $i \leq 10$

4. Print i

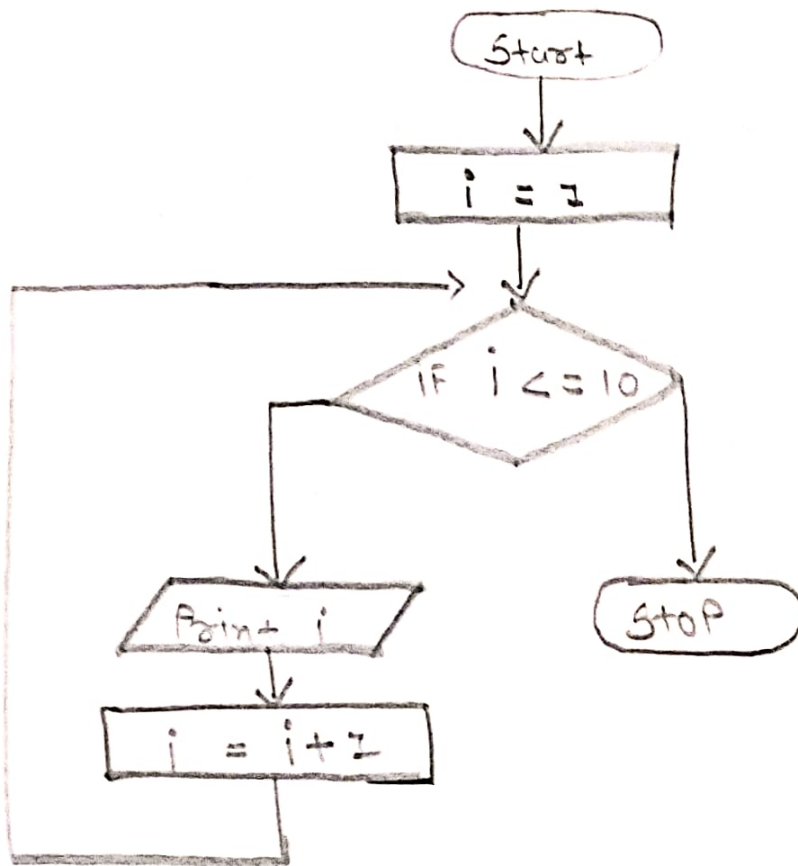
5. Increment $i = i + 1$

6. Repeat Steps 3 to 5 Condition become false

7. Stop.

Teacher's Signature: _____

★ Flowchart :-



EXPT. NO.	NAME	M T W T F S S	Page No.	YOUVA
			Date	

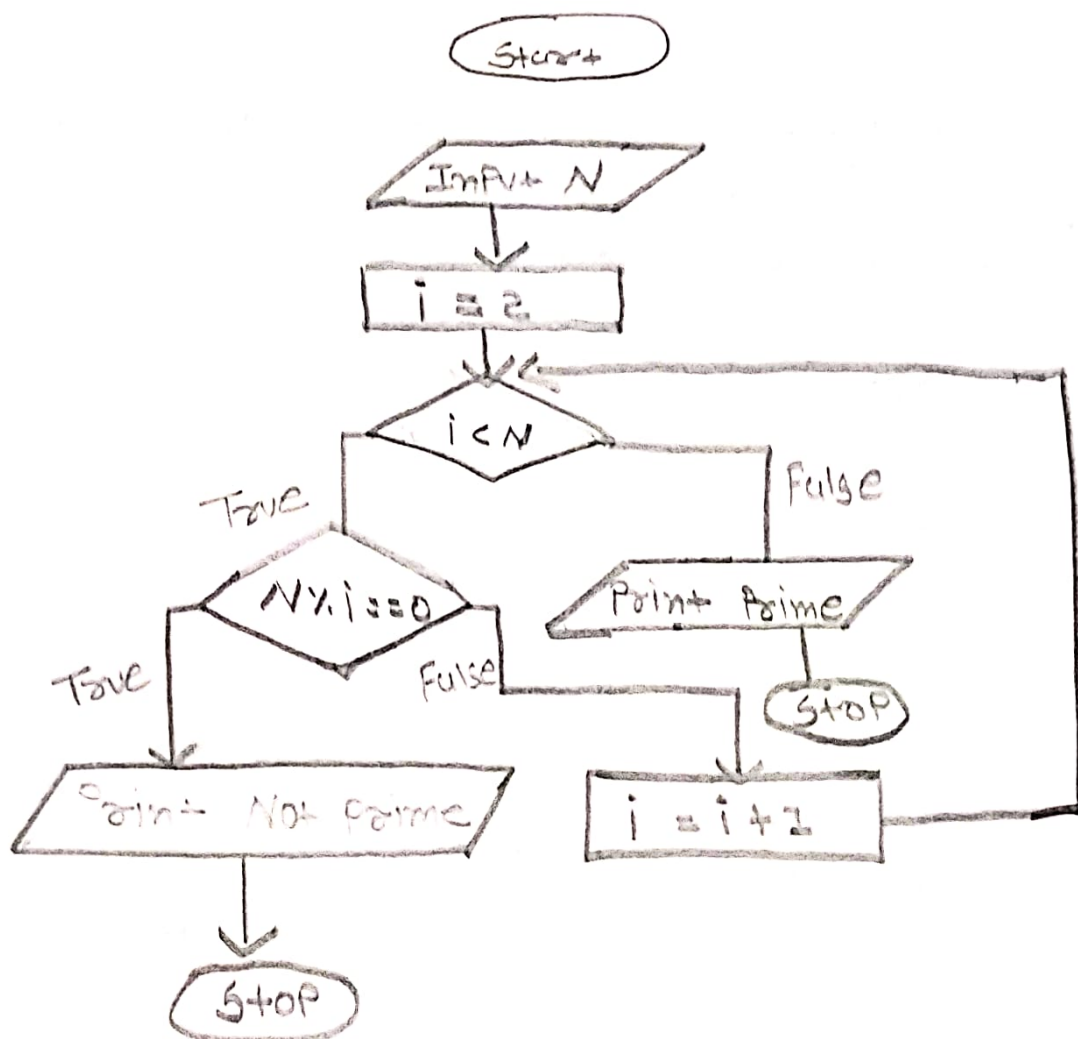
b) Write a Program to Show whether the entered number is prime or not.
(Flowchart & Algorithm)

→ Algorithm

1. Start
2. Input Number N
3. Set $i = 2$
4. Check if $N \% i == 0$
5. If true Print "Number is Not prime"
6. Otherwise increment $i = i + 1$
7. Repeat until $i < N$
8. if no divisor is found, Print "Number is Prime"
9. Stop.

Teacher's Signature: _____

★ Flowchart



5

★ Develop a program to demonstrate the basic functionalities of a Standard Calculator.

→ Program:-

```
def add (x,y):  
    return x+y
```

```
def Subtract (x,y):  
    return x-y
```

```
def multiply (x,y):  
    return x*y
```

```
def divide (x,y):  
    return x/y
```

```
Print (" --- Standard Calculator --- ")
```

```
Print (" Option : 1. Add, 2. Subtract, 3. Multiply,  
4. Divide")
```

```
Choice = int (input ("Enter your choice: "))
```

```
Num1 = Float (input ("Enter First Number: "))
```

```
Num2 = Float (input ("Enter Second Number: "))
```

★ Output :-

--- Standard Calculator ---

Option :

1. Add
2. Subtract
3. Multiply
4. Divide

Enter your choice : 1

Enter First Number : 10

Enter Second Number : 5

Result = 15


```
if Choice == 1:
    Print ("Result = ", Add(num1, num2))

elif Choice == 2:
    Print ("Result = ", Subtract(num1, num2))

elif Choice == 3:
    Print ("Result = ", Multiply(num1, num2))

elif Choice == 4:
    if num2 != 0:
        Print ("Result = ", divide(num1, num2))
    else:
        Print ("Division by zero is not possible")

else:
    Print ("Invalid Choice")
```

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A single, short, diagonal handwritten stroke or mark.

7. Write a Program to display the Following Patterns.

→ A)

```

1
1 2
1 2 3
1 2 3 4
1 2 3 4 5

```

Program :

```

for i in range (1,6):
    for j in range (1,1+i):
        print (j, end = " ")
    print ()

```

★ Output :-

1

1 2

1 2 3

1 2 3 4

1 2 3 4 5

B)

```

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★ ★ ★
★ ★
★

```

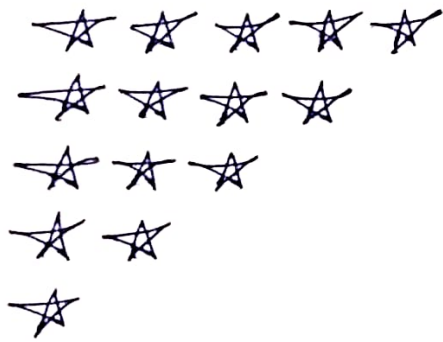
→ Program :-

```

for i in range (5, 0, -1):
    for j in range (0, i):
        print ("★", end = " ")
    print ()

```


★ Output :-



★ Flowchart & Algorithm

a) 1

1 2

1 2 3

1 2 3 4

1 2 3 4 5

→ Algorithm

1. Start

2. Initialize $i = 1$

3. Check Condition $i \leq 5$

4. Initialize $j = 1$

5. Check Condition $j \leq i$

6. Print j

7. Increment $j = j + 1$

8. Repeat steps 5 to 7 until Condition becomes False

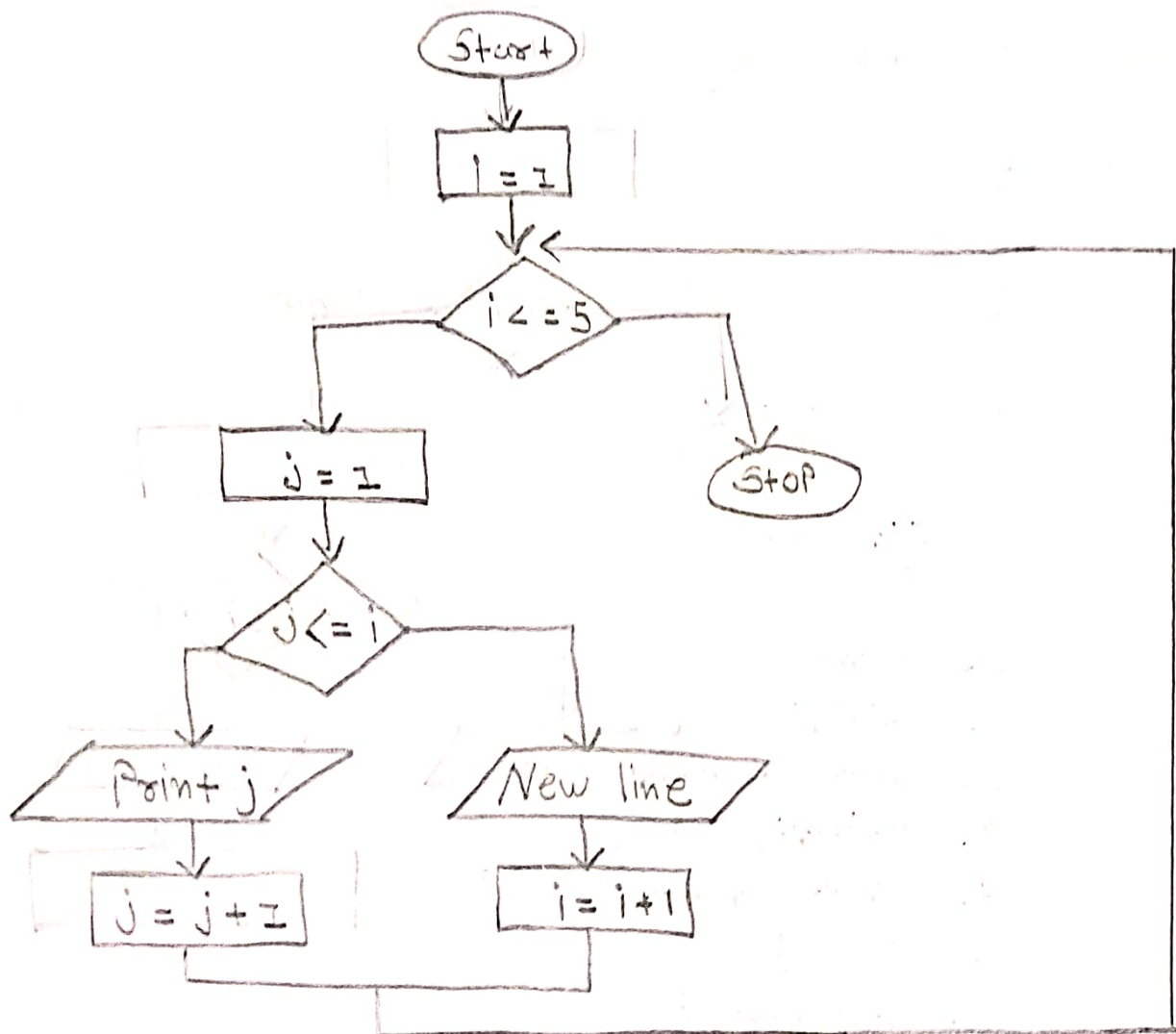
9. Print new line

10. Increment $i = i + 1$

11. Repeat steps 3 to 10 until Condition becomes False

12. Stop.

* Flowchart :-

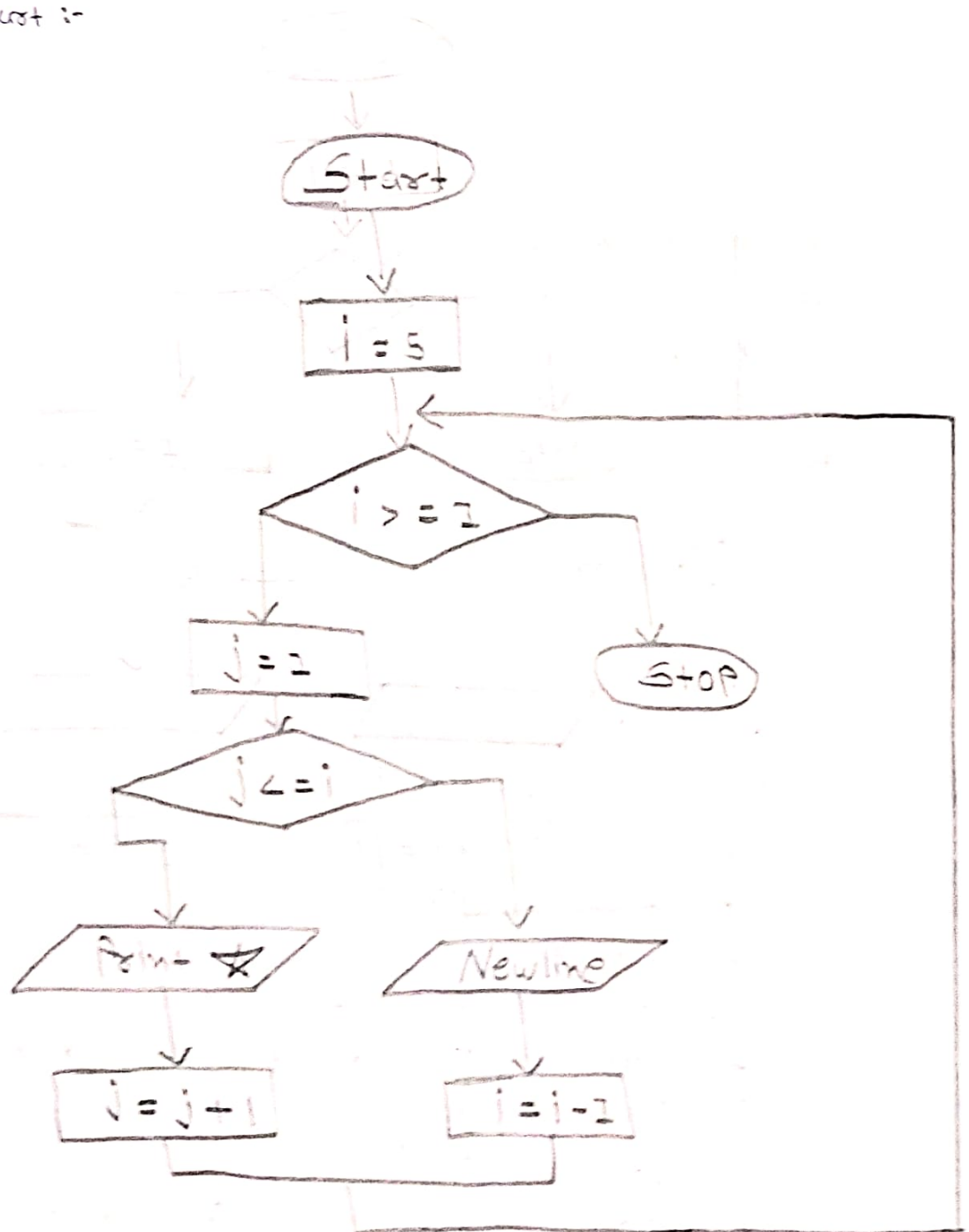


b) ★ ★ ★ ★ ★
 ★ ★ ★ ★
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→ Algorithm

1. Start
2. Initialize $i = 5$
3. Check Condition $i \geq 1$
4. Initialize $j = 1$
5. Check Condition $j \leq i$
6. Print ★
7. Increment $j = j + 1$
8. Repeat Step 5 to 7 until condition becomes False
9. Print new line
10. Decrement $i = i - 1$
11. Repeat Steps 3 to until Condition becomes False
12. Stop.

★ Flowchart :-



8. Create a User defined Function to print the Fibonacci series of 0 to N numbers.
(Where N is an integer number and passed as an argument).

→ Program :-

```
nterms = int(input("How many terms? "))
```

```
n1, n2 = 0, 1
```

```
Count = 0
```

```
if nterms <= 0:
```

```
    Print("Please enter a positive integer")
```

```
elif nterms == 1:
```

```
    Print("Fibonacci sequence upto", nterms, ":")
```

```
    Print(n2)
```

```
else:
```

```
    Print("Fibonacci sequence :")
```

```
    While Count < nterms:
```

```
        Print(n2)
```

```
        nth = n1 + n2
```

```
        n1 = n2
```

```
        n2 = nth
```

```
        Count += 1
```

★ Output :-

How many terms? 5

Fibonacci sequence :

0

1

1

2

3

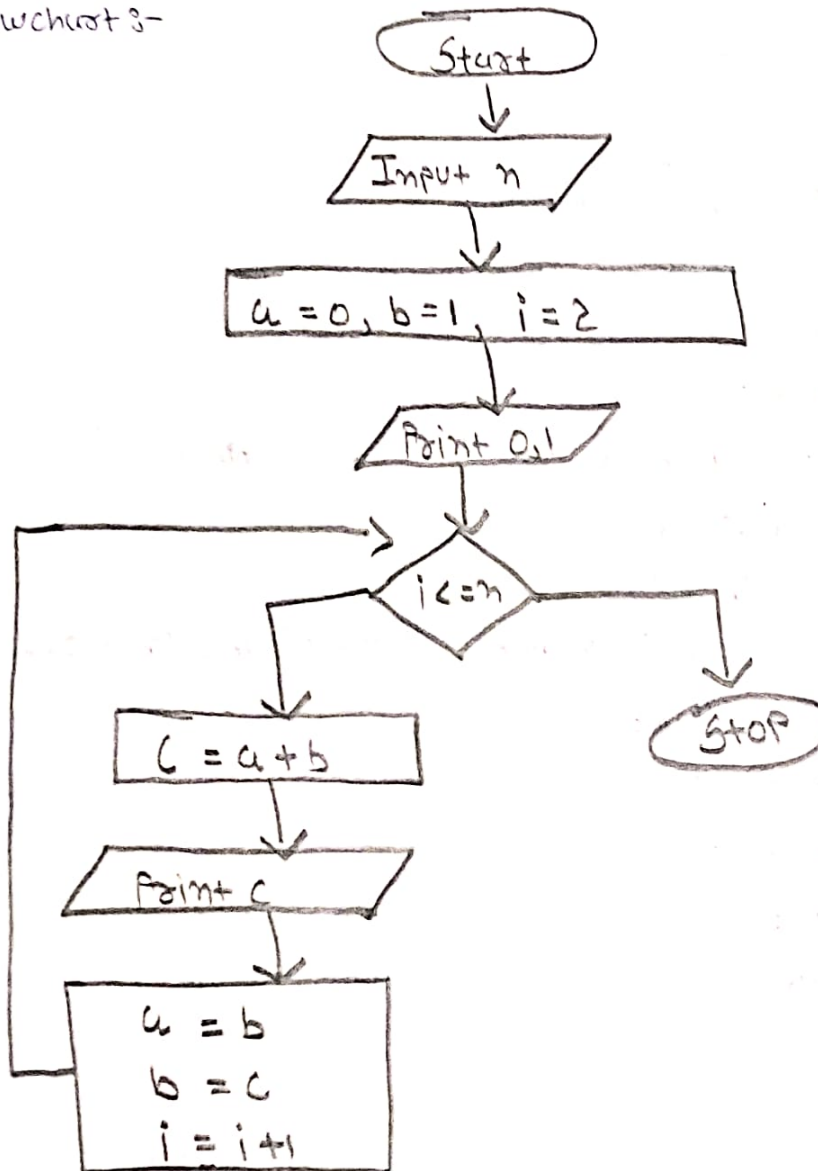
EXPT. NO.	NAME	M T W T F S S	Page No.	YOUVA
		Date		

→ Algorithm & Flowchart

1. Start
2. Input n
3. Define user defined function Fibonacci (n)
4. Initialize $a = 0$ and $b = 1$
5. Print a and b
6. Repeat until required terms are printed:
 - $c = a + b$
 - Print c
 - $a = b$
 - $b = c$
7. Stop

Teacher's Signature: _____

★ Flowchart 3-



EXPT. NO.	NAME	M T W T F S S	YOUVA
		Page No.:	
		Date	

Q.

a) Develop a User-defined Function to Find the Factorial of a given number.

→ Program

```
def factorial (num):
```

```
    if num == 1
```

```
        return 1
```

```
    else:
```

```
        return num * factorial (num-1)
```

```
num = int (input ("Enter a number: "))
```

```
result = factorial (num)
```

```
print ("The Factorial of ", num, "is", result)
```

Teacher's Signature: _____

★ Output :-

Enter a number : 6

The Factorial of 6 is 720

EXPT. NO.	NAME	M T W T F S S	YOUVA
		Page No.	
		Date	

b) Write a Program Using the Function that reverses the entered value.

Program :-

```
number = int(input("Enter the integer number: "))
```

```
revs = num = 0
```

```
while (number > 0):
```

```
    remainder = number % 10
```

```
    revs = num * 10 + remainder
```

```
    number = number // 10
```

```
print("The reverse number is: {}".format(revs - num))
```

Teacher's Signature: _____

★ Output :-

Enter the integer number : 2484

The reverse number is : 4842



EXPT. NO.	NAME	M T W T F S S	YOUVA
		Page No.	
		Date:	

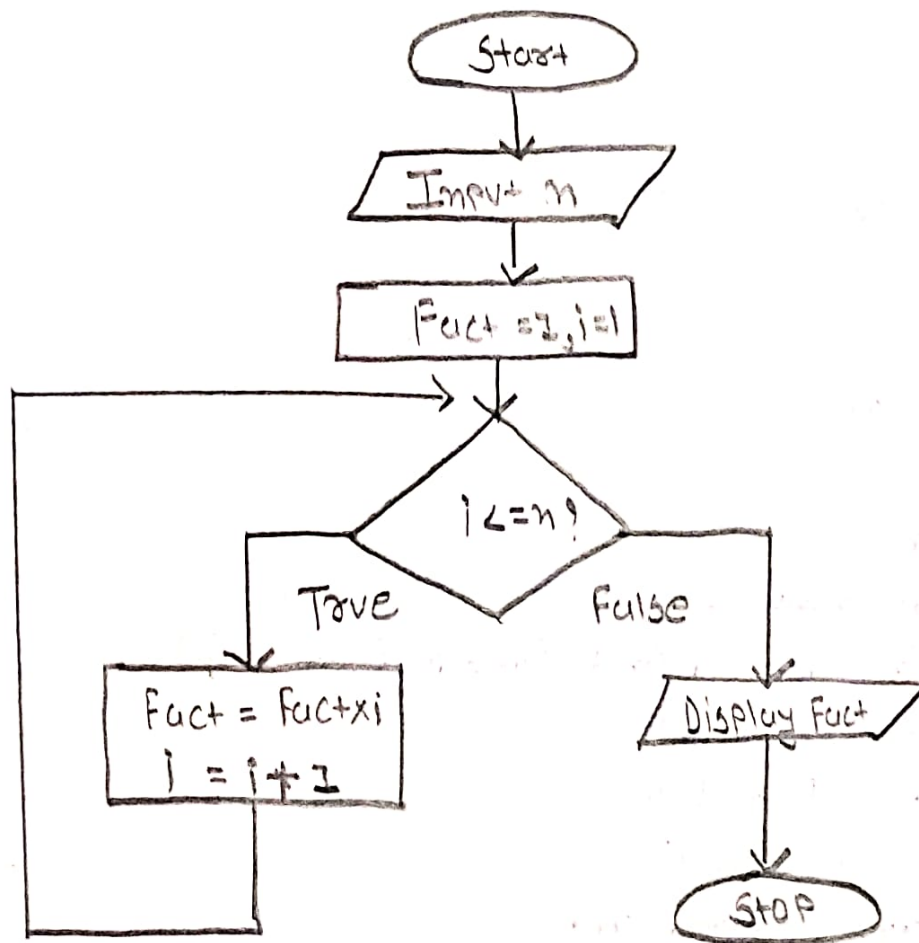
9. Algorithm & Flowchart

(a)

Algorithm

1. Start
2. input number n
3. Define User defined Function Factorial (n)
4. Initialize $fact = 1$
5. Repeat From 1 to n
 - $fact = fact \times i$
6. Return Factorial value
7. Display Factorial
8. stop

Teacher's Signature: _____



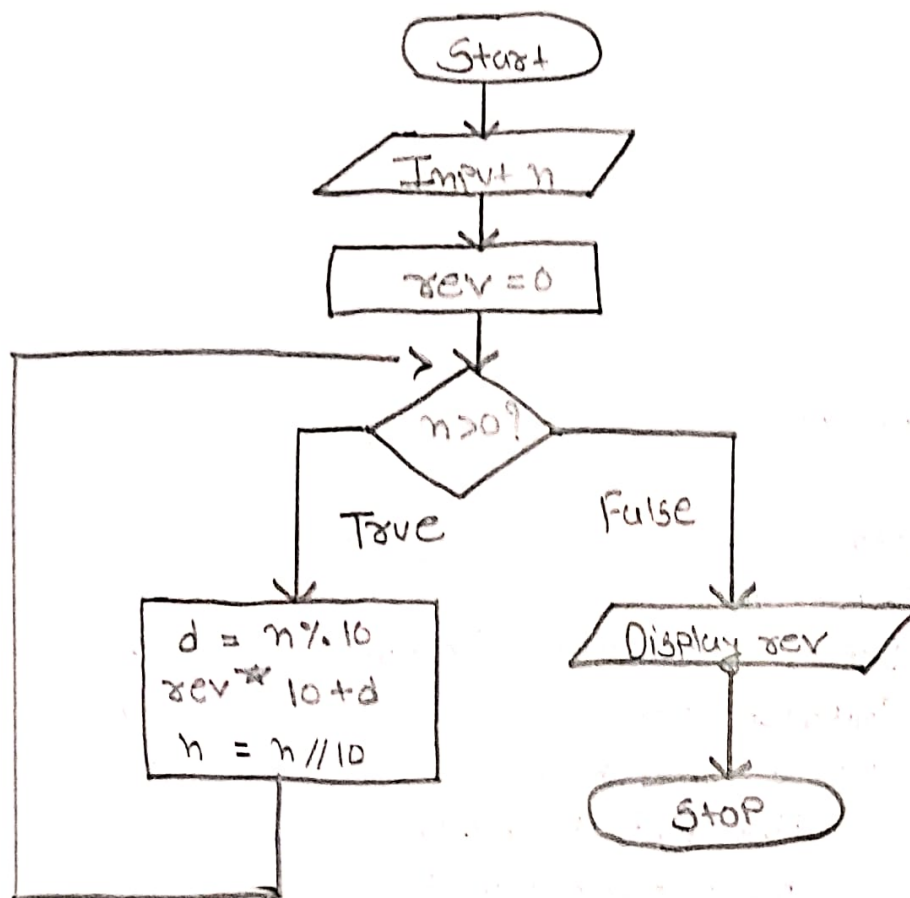
EXPT. NO.	NAME	M T W T F S S	YOUVA
		Page No.	
		Date	

b)

Algorithm

- 1) Start
- 2) Input number n
- 3) Define user defined Function $\text{reverse}(n)$
- 4) Initialize $\text{rev} = 0$
- 5) Repeat while $n > 0$
 - Find digit $d = n \% 10$
 - $\text{rev} = \text{rev} \times 10 + d$
 - $n = n // 10$
- 6) Return reversed number
- 7) Display reversed number
- 8) Stop

Teacher's Signature: _____



10)

a) Write a Program that determines whether a given number is an 'Armstrong number' or a Palindrome Using a user-defined function.

→ Program :-

```
def Check - Armstrong (num):
    num - str = str (num)
    n = len (num - str)
    arm - sum = sum (int (digit) ** n for
                      digit in num - str)
    return arm - sum == num
```

```
def Check - Palindrome (num):-
    num - str = str (num)
    return num - str == num - str [::-1]
```

try:

```
user - input = int (input ("Enter a number: "))
```

```
if Check - Armstrong (user - input):
    print (f" {user - input} is an Armstrong
           number. ")
```

★ Output :-

Enter a number : 6

6 is ~~an~~ an Armstrong number.

6 is a Palindrome.

EXPT. NO.	NAME	M T W T F S S	YOUVA
		Page No.	
		Date	

else:•

Print (f" {user - input} is NOT an Armstrong number.")

if check_Palindrome (user - input):

Print (f" {user - input} is a Palindrome. ")

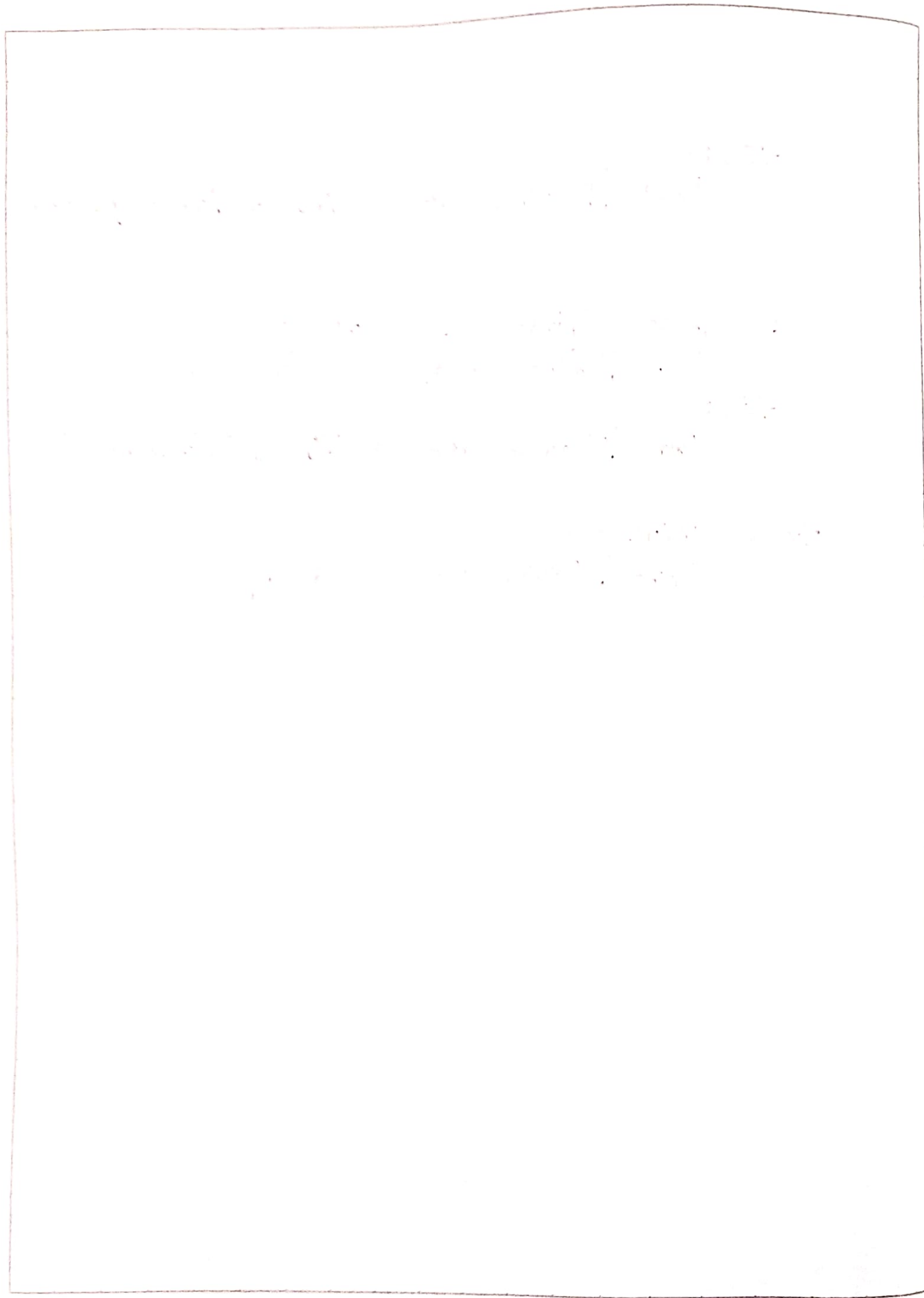
else:

Print (f" {user - input} is NOT a Palindrome. ")

Except Value Error:

Print ("Please enter a valid integer. ")

Teacher's Signature: _____



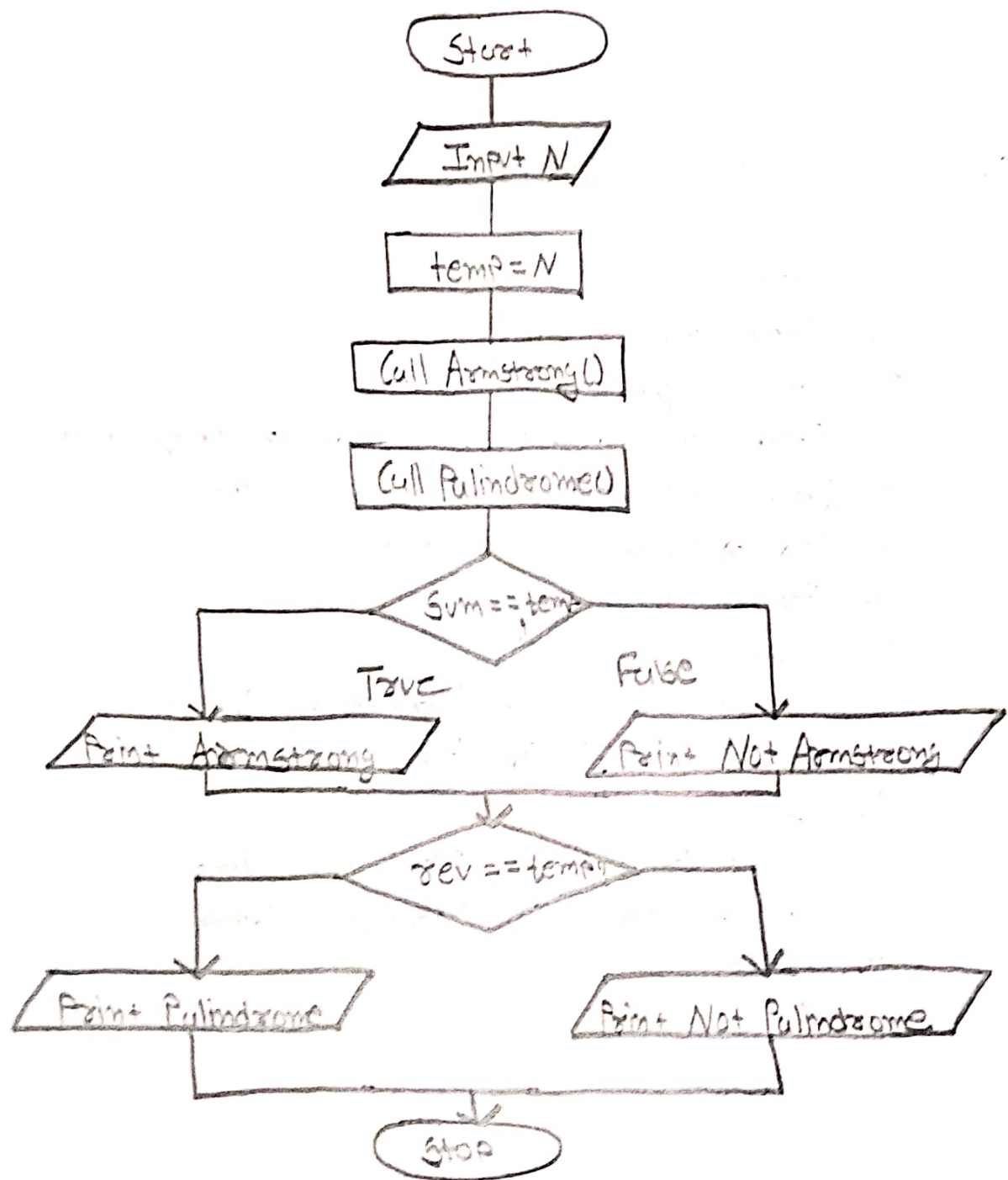
EXPT. NO.	NAME	M T W T F S S	Page No.	YOUVA
			Date	

10)

a) Algorithm & Flowchart

1. Start
2. Input number N
3. Store original number in temp
4. Define user defined function for Armstrong check
5. Define user defined function for Palindrome check
6. Find Armstrong sum of digits
7. Reverse the number
8. Check $sum == temp$
 - If true, Print "Armstrong Number"
 - Else Print "Not Armstrong Number"
9. Check $reverse == temp$
 - If true, Print "Palindrome Number"
 - Else Print "Not Palindrome Number"
10. Stop

Teacher's Signature _____



EXPT NO.	NAME	Page No.	DATE
			YOUVA

11) Algorithm & Flowchart

(a) Write a Program to Find the length of String

→ Algorithm

1. Start

2. Input a String str

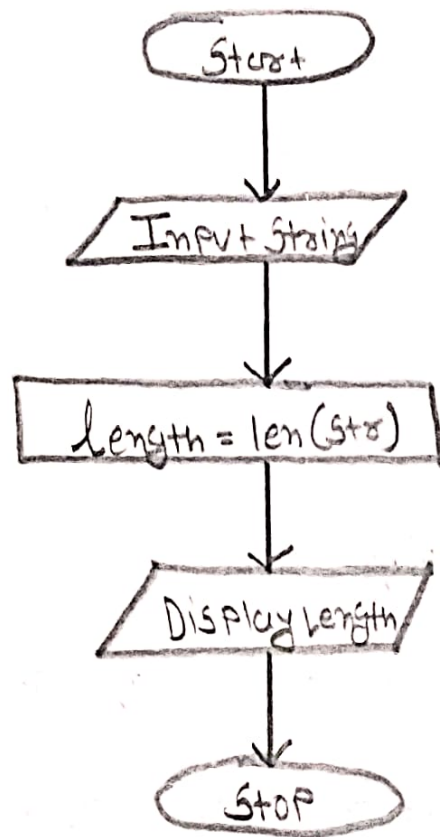
3. Find length of string using len() function

4. Store result in variable length

5. Display length

6. Stop

Teacher's Signature



EXPT. NO.	NAME	M T W T F S S	YOUVA
		Page No.	
		Date	

b) Write a Program to reverse words in a given Sentence.

→ Algorithm

1. Start

2. Input a Sentence

3. Split Sentence into words

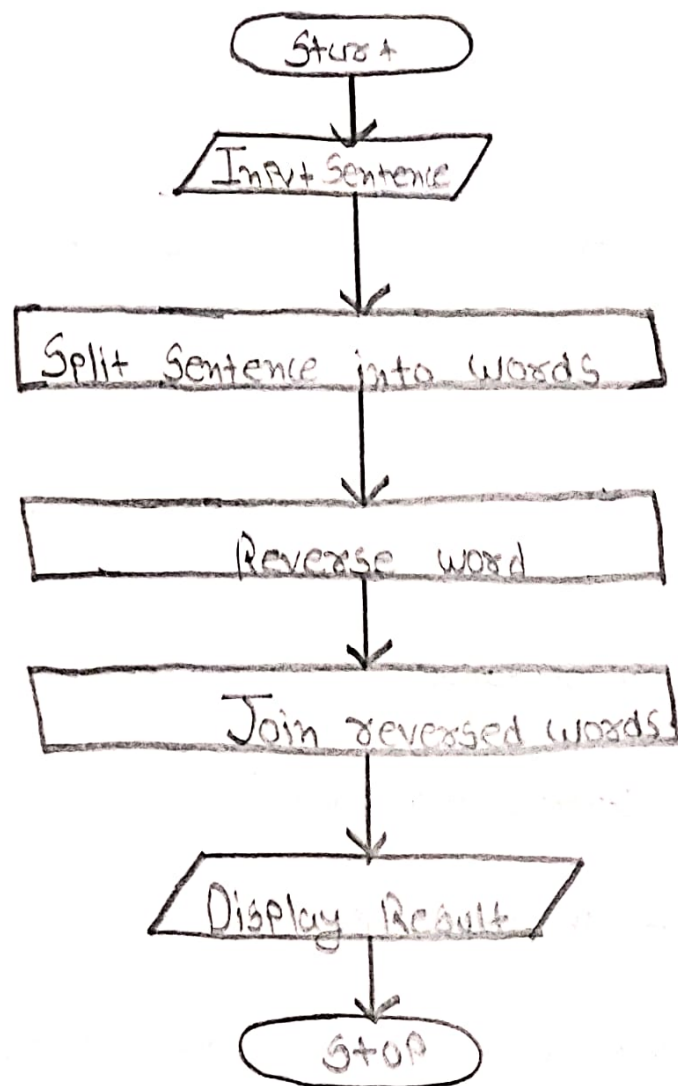
4. Reverse the Order OF words

5. Join reversed word into a new sentence

6. Display reversed Sentence

7. Stop

Teacher's Signature: _____



EXPT. NO.	NAME	M T W T F S S	YOUVA
		Page No.	
		Date	

C) Write a program to check if a substring is present in a given string.

→ Algorithm

1. Start

2. Input main string str

3. Input substring sub

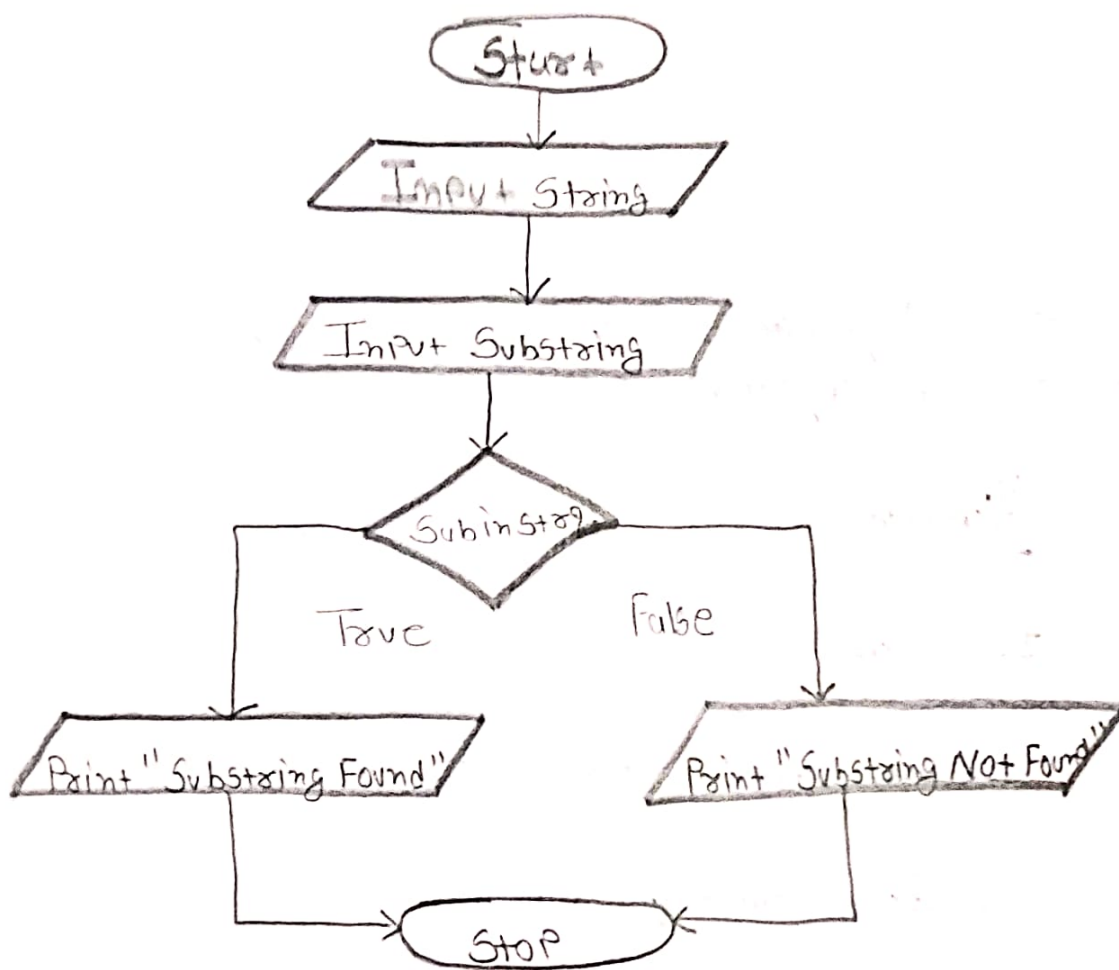
4. Check whether sub is present in str.

5. If present, display "Substring Found"

6. Otherwise, display "Substring Not Found"

7. Stop

Teacher's Signature: _____



EXPT. NO.	NAME	M T W T F S S	YOUVA
		Page No.	
		Date	

d) Write a program to Count and display the number of vowels, Consonants, uppercase, lowercase, Characters in a String.

→ Algorithm

1. Start

2. Input a String

3. Initialize Counters:

- Vowel = 0
- Consonant = 0
- uppercase = 0
- lowercase = 0

4. Read each Character OF String

5. Check:

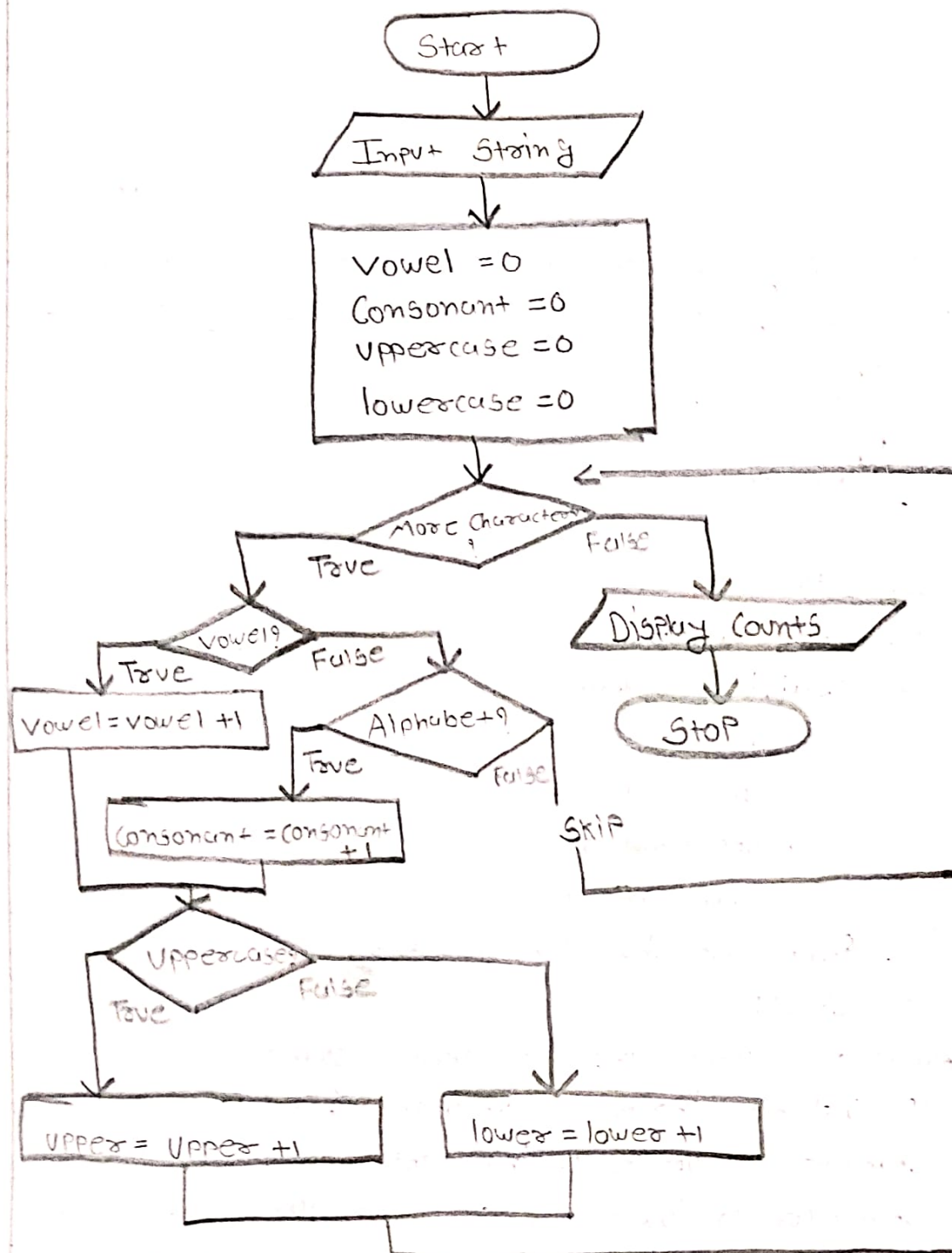
- if Character is vowel, increment vowel counter
- Else if Alphabet, increment Consonant Counter
- If Character is uppercase, increment uppercase Counter
- If Character is lowercase, increment lowercase Counter

6. Repeat until all characters are checked

7. Display all Counts

8. STOP

Teacher's Signature: _____



EXPT. NO.	NAME	M T W T F S S	Page No.	YOUVA
		Date		

12) Algorithm & Flowchart

a) To swap given two element in a list.

→ Algorithm

1. Start

2. Input list elements

3. Input Positions / elements to swap (A, B)

4. Store First element in temporary variable temp

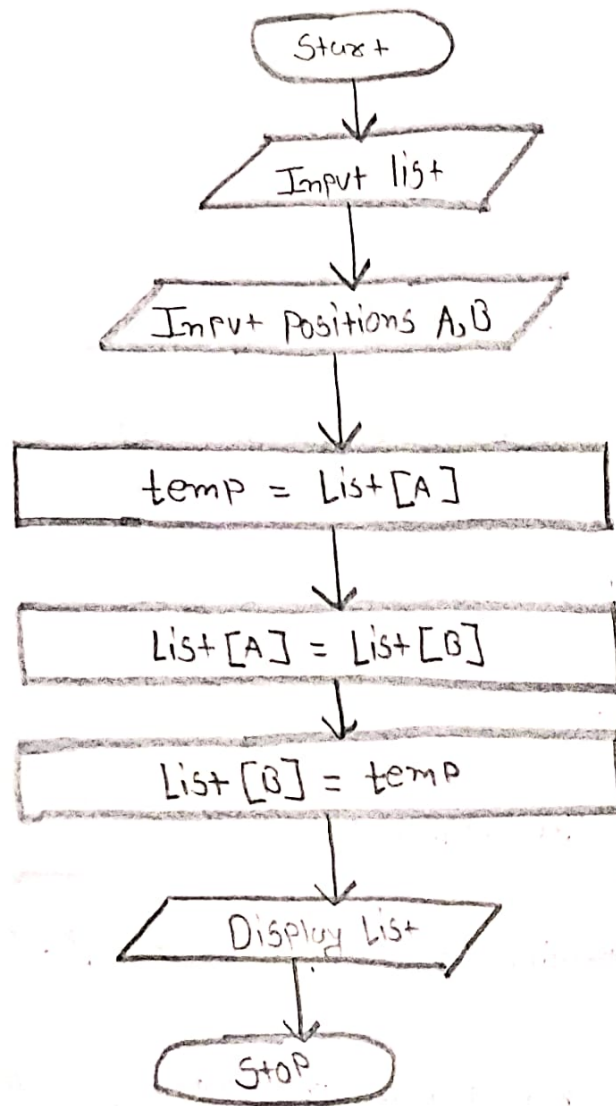
5. Assign second element to First position

6. Assign temp value to second position

7. Display updated list

8. Stop

Teacher's Signature: _____



EXPT. NO.	NAME	M T W T F S S	Page No.:	YOUVA
		Date:		

b) To find the sum of elements in a list.

→ Algorithm .

1. Start

2. Input list elements

3. Initialize $sum = 0$

4. Read each element from the list

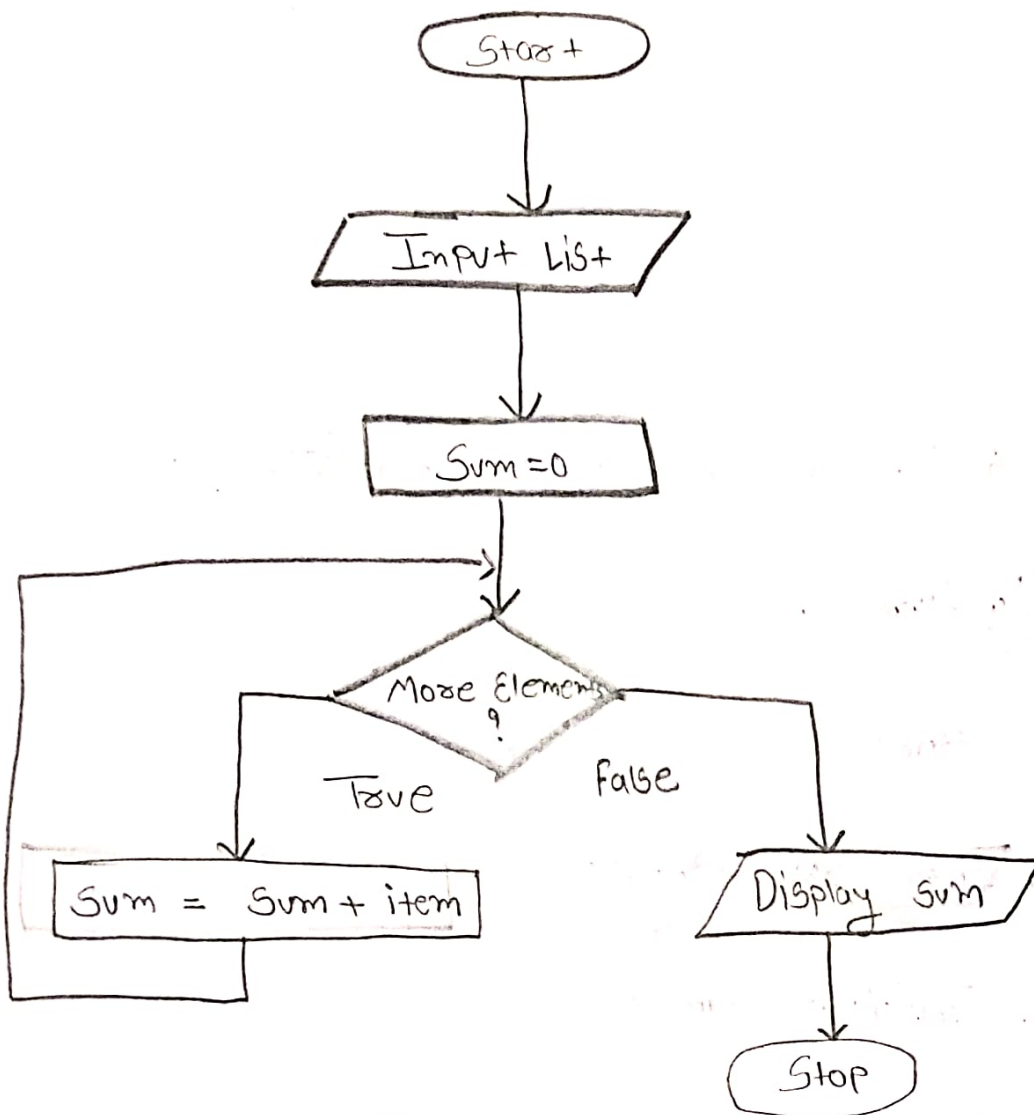
5. Add element to sum

6. Repeat until all elements are added

7. Display total sum

8. Stop.

Teacher's Signature: _____



EXPT. NO.	NAME	M T W T F S S	Page No.:	YOUVA
		Date:		

c) To Check if an element exists in a given list.

→ Algorithm

1. Start

2. Input list elements

3. Input element to search

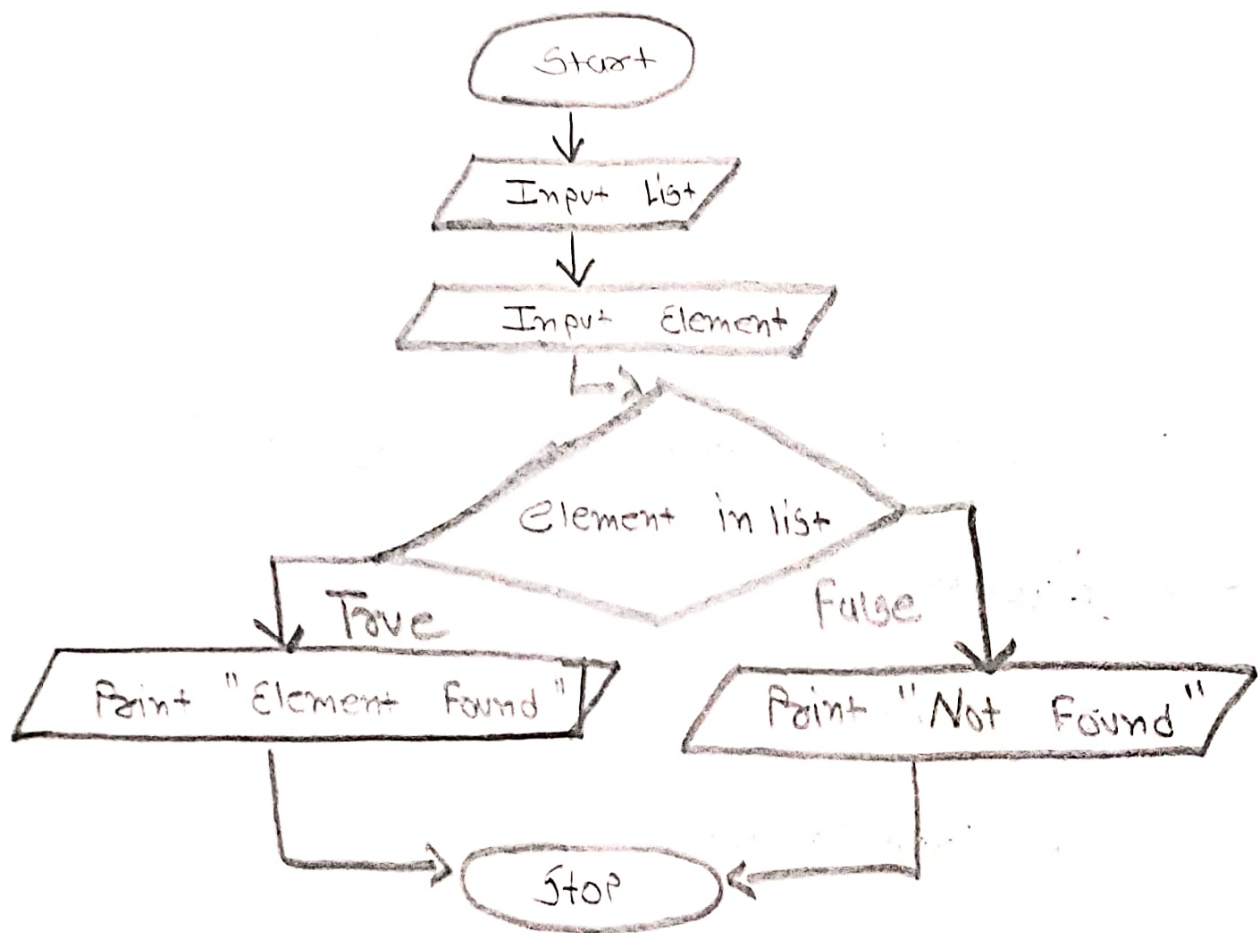
4. Check whether element exists in the list

5. IF found, display "Element found"

6. Otherwise, display "Element Not Found"

7. Stop

Teacher's Signature: _____



d) To Find the smallest and largest element in a given list.

→ Algorithm

1. Start

2. Input list elements

3. Assign First element as:

- smallest
- largest

4. Compare each element in the list

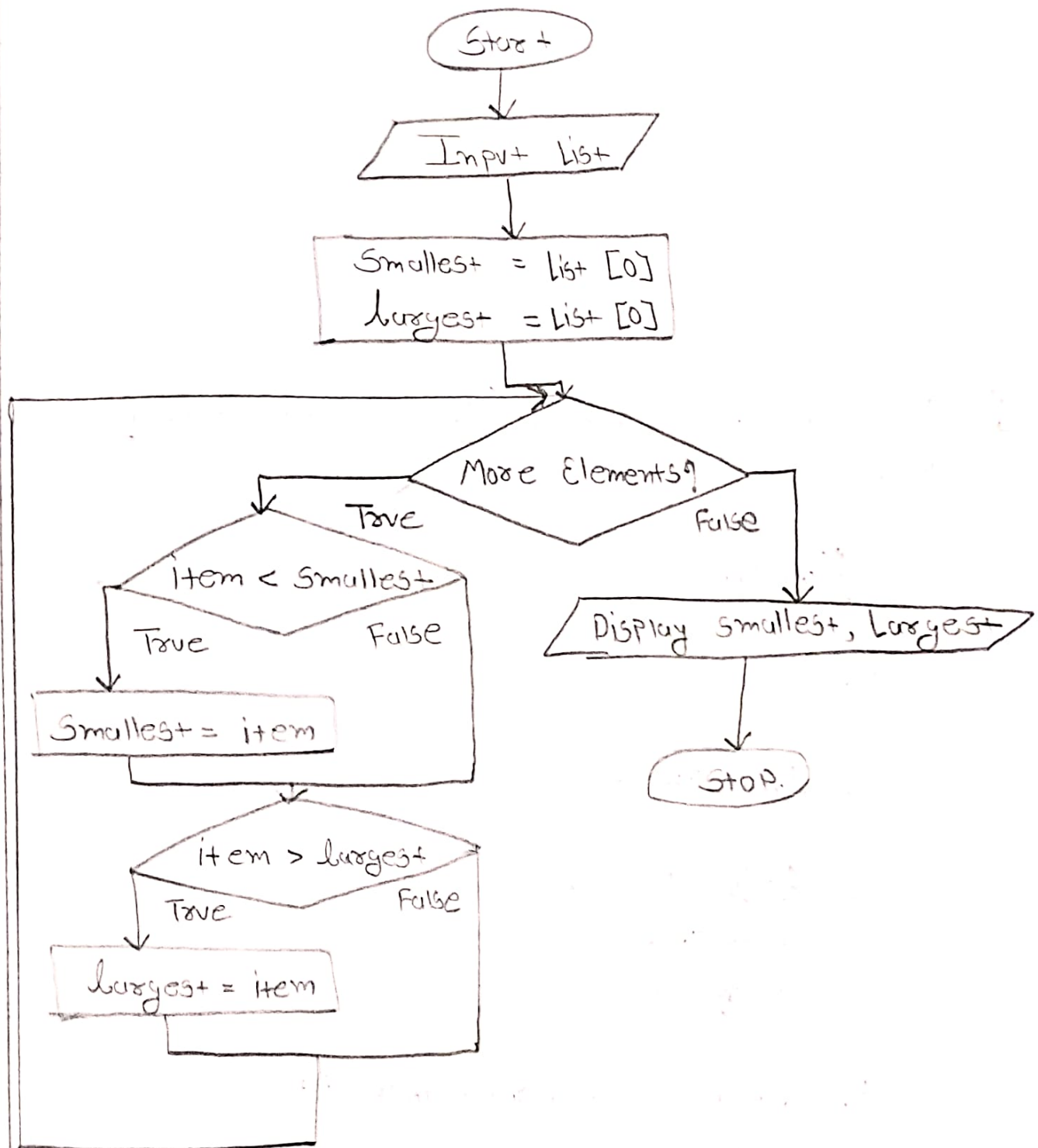
5. If element $<$ smallest, update smallest

6. If element $>$ largest, update largest

7. Repeat until all elements are checked

8. Display smallest and largest element

9. stop.

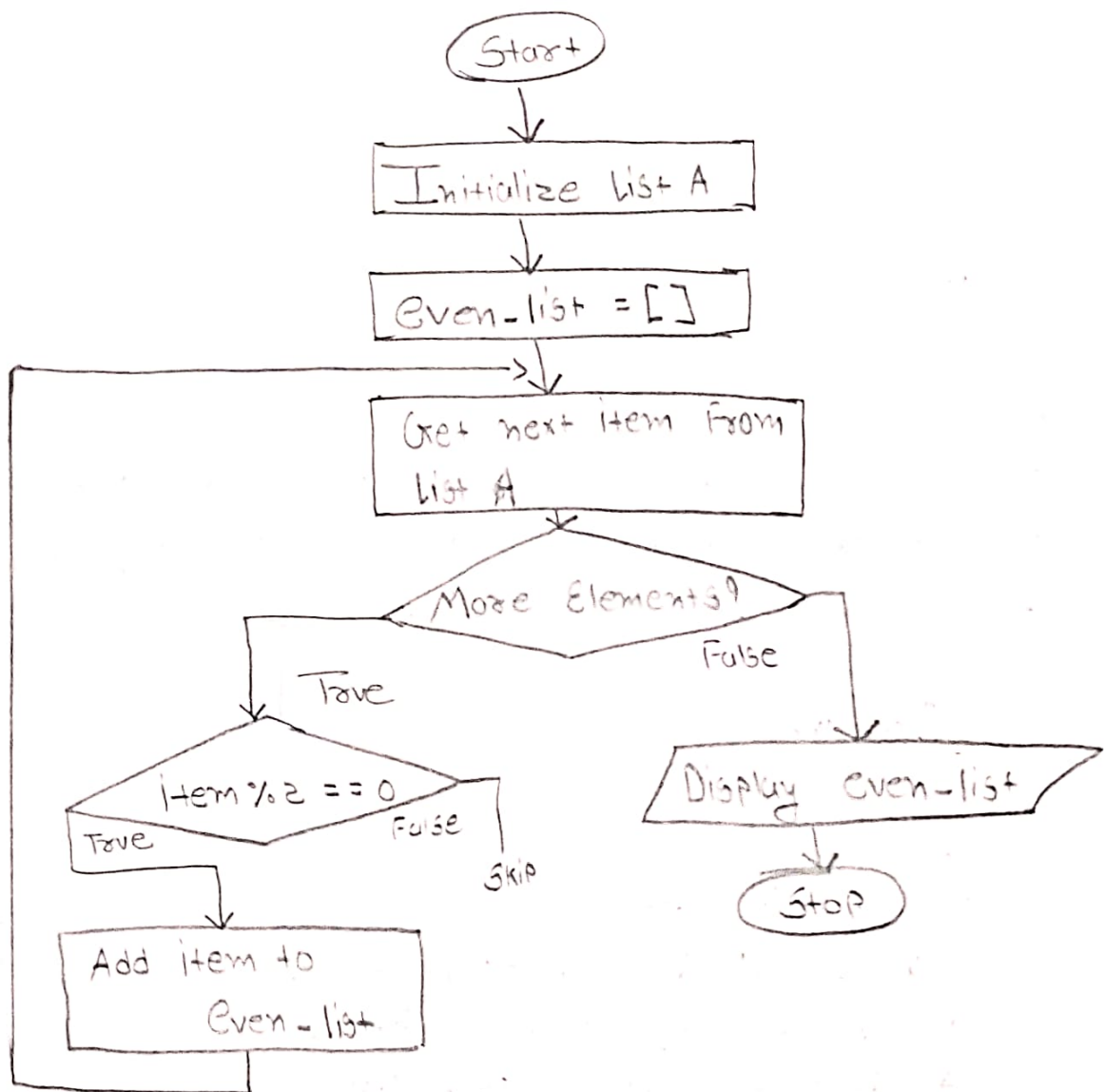


13) Algorithm & Flowchart

a) Given a list saved in var: $a = [1, 4, 9, 16, 25, 36, 49, 64, 81, 100]$. Write one line of Python that takes this list and makes a new list that has only the even elements of this list in it.

→ Algorithm

1. Start
2. Initialize list
 $a = [1, 4, 9, 16, 25, 36, 49, 64, 81, 100]$
3. Create empty list even-list = []
4. Read each element from list a
5. Check if element is even
6. If even, add element to even-list
7. Repeat until all elements are checked
8. Display even-list
9. Stop



EXPT. NO.	NAME	M T W T F S S	YOUVA
		Page No.	
		Date	

b) Create a list containing the square of all odd numbers from range 1 to 10.

→ Algorithm

1. Start

2. Create empty list `square-list = []`

3. Initialize number from 1-10

4. Check whether number is odd

5. If odd, find square of number

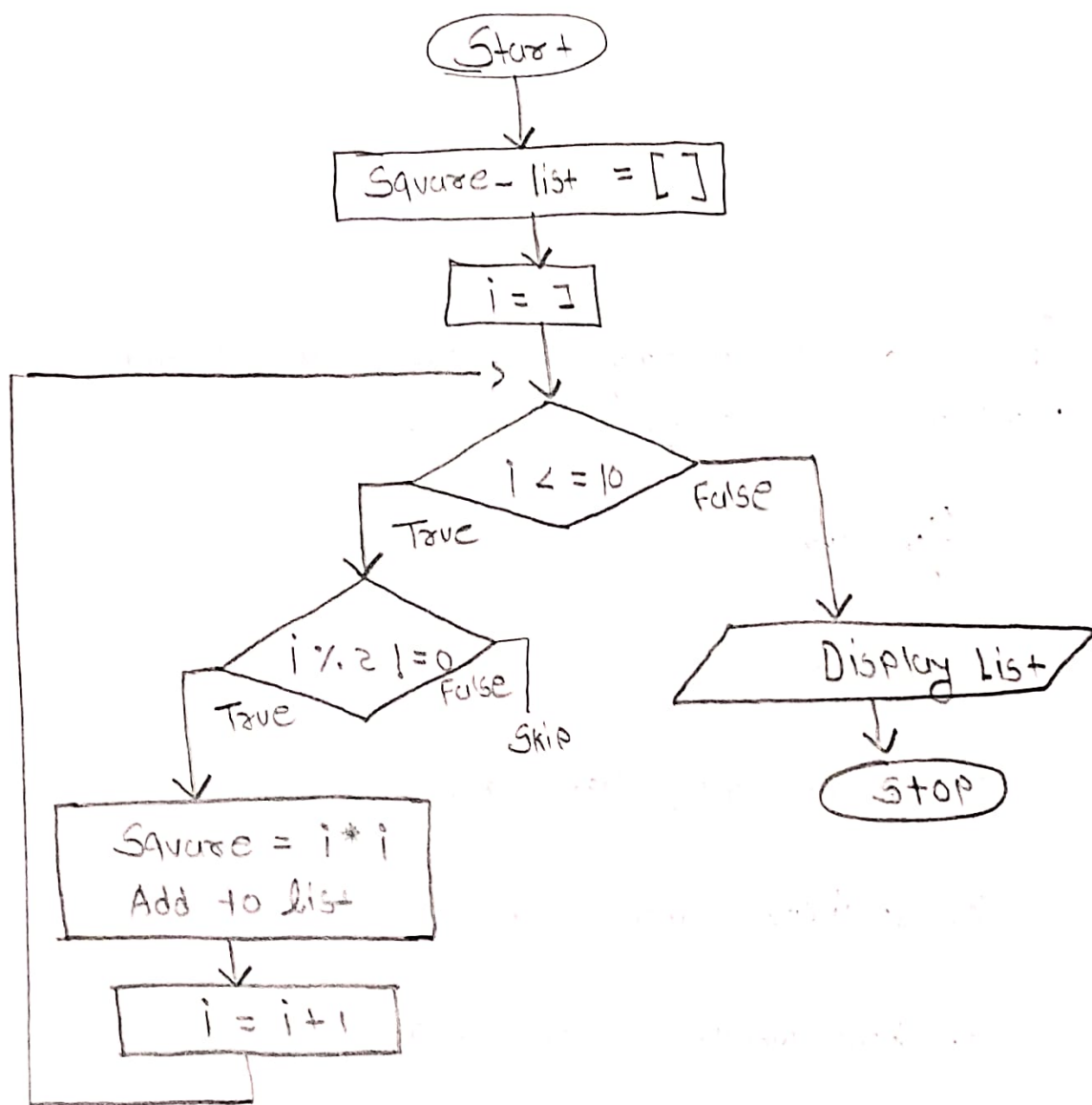
6. Add square to `square-list`

7. Repeat until all numbers are checked

8. Display `square-list`

9. stop

Teacher's Signature: _____



EXPT. NO.	NAME	M T W T F S S	YOUVA
		Page No.	
		Date	

C) Create a list of Prime and non-Prime numbers in range 1 to 50.

→ Algorithm

1. Start

2. Create empty list:

• Prime_list = []

• nonprime_list = []

3. Take numbers from 1-50

4. For each number, check divisibility

5. If Number has only two factors (1 and itself),

add to Prime_list

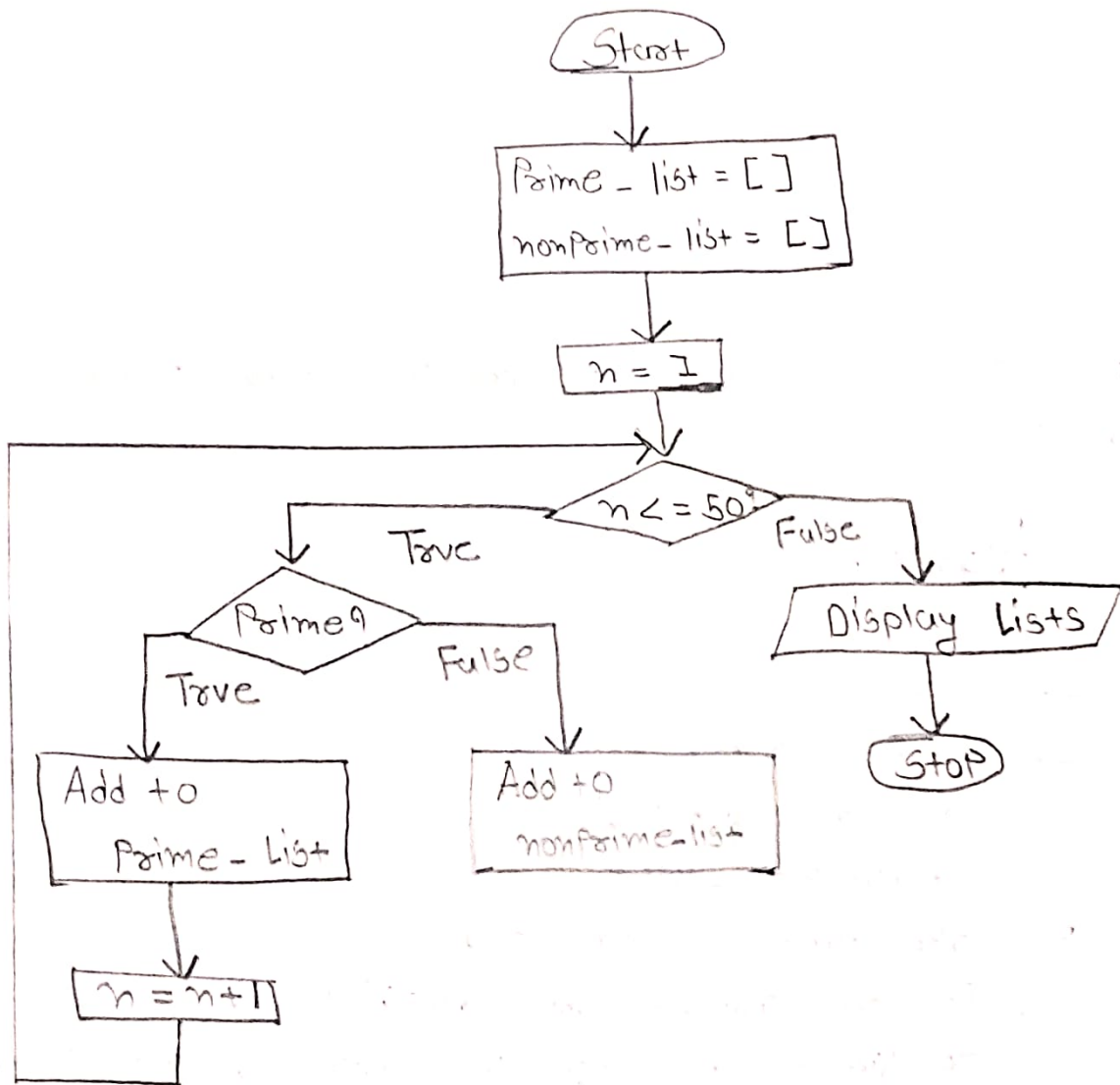
6. Otherwise add to nonprime_list

7. Repeat until all numbers are checked

8. Display Prime list and non-Primelist.

9. Stop

Teacher's Signature: _____



EXPT. NO.	NAME	M T W T F S S						
		Page No. _____ Date: _____						

YOUVA

~~*/~~ To Find the smallest and largest element in a given list.

Teacher's Signature: _____

EXPT. NO.	NAME	M T W T F S S	YOUVA
		Page No.:	
		Date:	

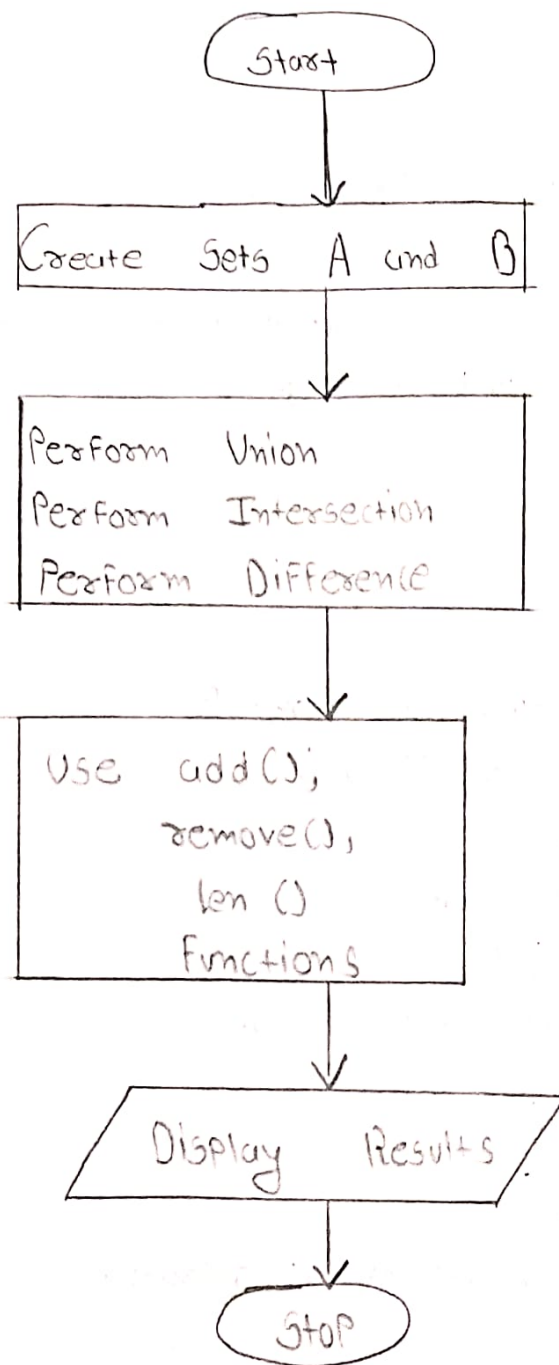
14. Algorithm & Flowchart

a) Write a program to demonstrate some useful set functions and operations.

→ Algorithm

1. Start
2. Create two sets A and B
3. Perform set operations:
 - Union
 - Intersection
 - Set Difference
4. Use useful set functions:
 - add()
 - remove()
 - len()
5. Display results of all operations
6. Stop

Teacher's Signature: _____



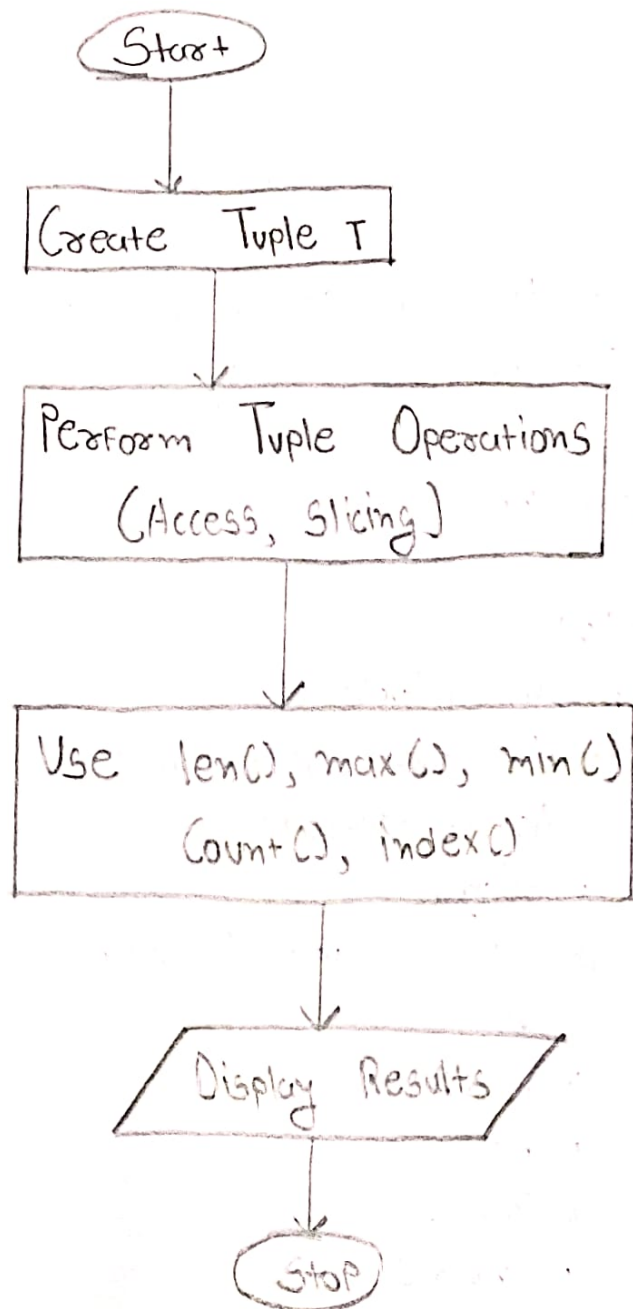
EXPT. NO.	NAME	M T W T F S S						
		Page No.:						
		Date:						
		YOUVA						

b) Write a program to demonstrate some useful Tuple Functions and Operations.

→ Algorithm

1. Start
2. Create a tuple T
3. Perform tuple Operations:
 - Access elements
 - Slicing
4. Use useful tuple Functions:
 - len()
 - max()
 - min()
 - count()
 - index()
5. Display results
6. Stop

Teacher's Signature: _____



EXPT. NO.	NAME	M T W T F S S	YOUVA
		Page No.	
		Date	

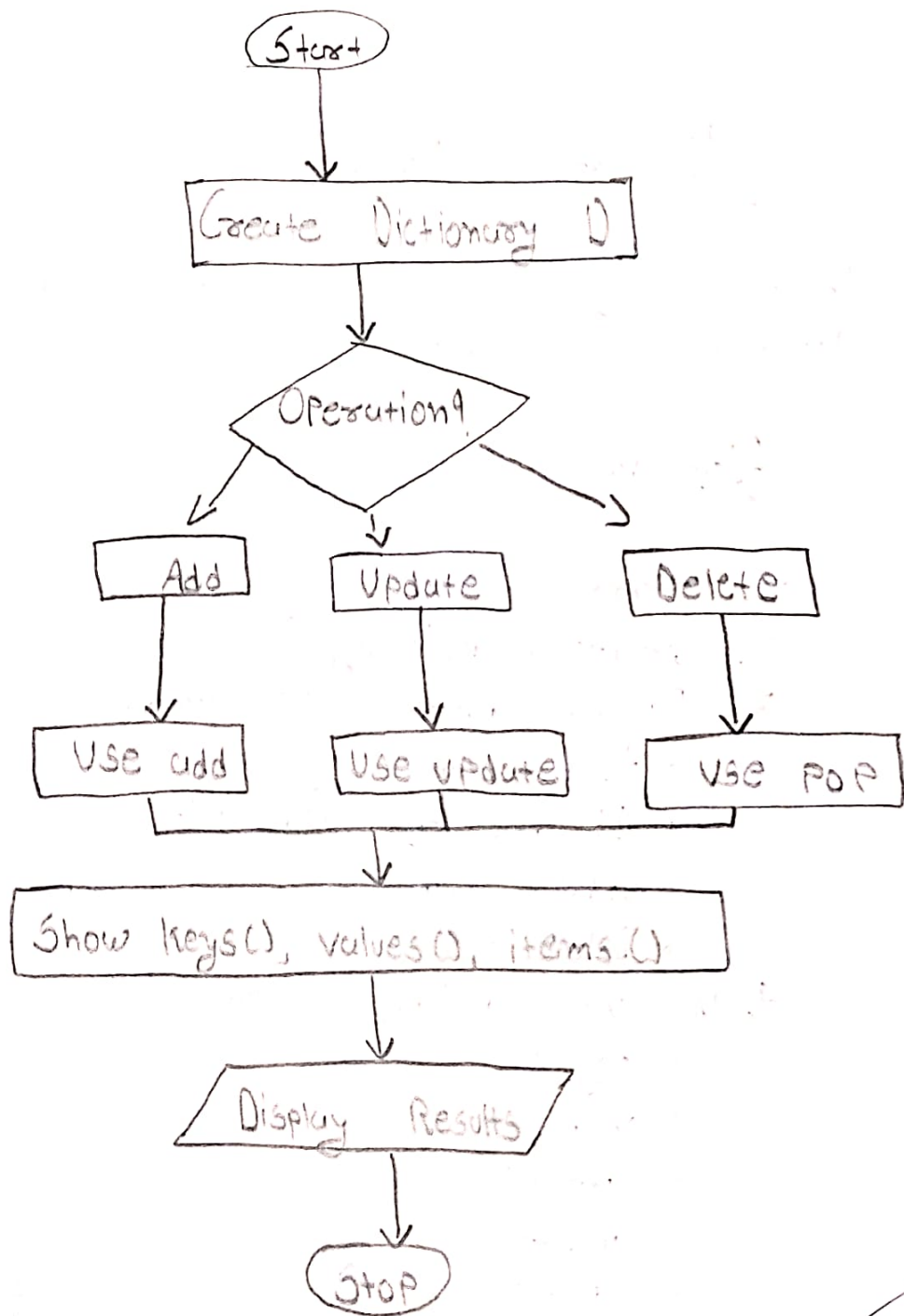
15. Algorithm & Flowchart

a) Write a Program to demonstrate the dictionaries Functions and Operations.

→ Algorithm

1. Start
2. Create a dictionary D
3. Perform dictionary Operations:
 - Access values using keys
 - Add new key - Value Pair
 - Update values
 - Delete element
4. Use useful dictionary Functions:
 - keys()
 - values()
 - items()
 - update()
 - pop()
5. Display results of all operations
6. Stop

Teacher's Signature: _____



EXPT. NO.	NAME	M T W T F S S	
		Page No.	YOUVA
		Date:	

b) Create a Dictionary with the roll number, name, and marks of n students in a class and display the names of students who have scored marks above 75.

→ Algorithm

1. Start
2. Input number of students n
3. Create empty dictionary D
4. Repeat for n students:
 - Input Roll Number
 - Input Name
 - Input Marks
 - Store details in dictionary
5. Read each student record from dictionary
6. Check if marks > 75
7. If true, display student name
8. Repeat until all records are checked
9. Stop

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